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Lignin fractionation from lignocellulosic biomass using deep eutectic solvents and its valorization

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ABSTRACT

Lignin, the most abundant natural aromatic feedstock, is one of the most renewable resources for many biofuels, chemicals, and smart materials. Recent progress in deep eutectic solvents (DESs), a class of green solvents exhibiting appealing properties, suggests promising pathways for lignin fractionation and further valorization. Despite considerable efforts that have been devoted to their development, the fractionation and valorization of lignin using DESs are still in their nascent stage. This review presents a comprehensive summary of recent progress in lignin fractionation and valorization using DESs. The introduction of emerging type V DESs, and the relationship between Kamlet–Taft solvatochromic parameters and lignin fractionation and solubility are emphasized. Key factors that affect the lignin fractionation yield and lignin structure have been identified. Particularly, recent growth in computation-based studies to determine which DESs are appropriate for lignin biorefinery and the investigation of underlying mechanisms are also highlighted. Furthermore, the properties of DESs-fractionated lignin and its further valorization routes are systematically summarized. Finally, future perspectives on the development of more efficient and sustainable lignin fractionation and valorization are provided. The present work will provide valuable guidance on the fractionation and valorization of lignin by using renewable DESs.

1. Introduction

Dwindling non-renewable fuel resource reserves and rapidly deteriorating environmental conditions have driven our society to search for clean and sustainable alternative energy and chemical sources. Lignocellulosic biomass is a renewable and abundant feedstock with great promise for the production of biofuels, chemicals, and materials [1]. Lignocellulosic biomass has many appealing merits: renewable, abundant, inexpensive, environmentally friendly, and does not compete with the feedstocks as a source of production. It is mainly composed of three components, i.e., cellulose, hemicellulose, and lignin [2]. Over the past decades, cellulose and hemicellulose have been successfully converted into a broad spectrum of biofuels and chemicals, such as ethanol, butanol and 5-hydroxymethylfurfural [3], which have achieved

Abbreviations: AA, acetic acid; AcChCl, acetylcholine chloride; ATACMC, allytrimethylammonium chloride; ATR-FTIR, attenuated total reflectance Fourier transform infrared; BzChCl, benzyldimethyl(2-hydroxyethyl)ammonium chloride; BTEAC, benzyltriehtylammonium chloride; BTMAC, benzyltrimethylammonium chloride; CEL, cellulolytic enzyme lignin; ChCl, choline chloride; COSMO-RS, conductor-like screening models for real solvents; DAL, dealkaline lignin; DBU, 1,5-diazabicyclo[5.4.0]-5-undecene; DESs, deep eutectic solvents; DFRC, derivatization followed by reductive cleavage; DSC, differential scanning calorimetry; DyP, dye-decolorizing peroxidase; EG, ethylene glycol; EHL, enzymatic hydrolysis lignin; EMAL, enzymatic/mild acidolysis lignin; FA, formic acid; GA, glutaric acid; GH, guanidine hydrochloride; GPC, gel permeation chromatography; HBA, hydrogen bond acceptor; HBD, hydrogen bond donor; K–T, Kamlet–Taft; LA, lactic acid; LigP, heme-dependent lignin peroxidase; MA, malonic acid; MD, molecular dynamics; MnP, manganese peroxidase; NBO, alkaline nitrobenzene oxidation; Mn, number-average molecular weight; M_W, weight-average molecular weight; NMR, nuclear magnetic resonance; OA, oxalic acid; OL, organosolv lignin; PB, *p*-hydroxybenzoic acid; PBA, *p*-hydroxybenzyl alcohol; PCA, *p*-coumaric acid; PDI, polydispersity index; PEG, polyethylene glycol; PHA, *p*-hydroxybenzaldehyde; Py-GC/MS, pyrolysis-gas chromatography-mass spectrometry; QC, quantum chemistry; SEC, size exclusion chromatography; SL, sodium lignosulfonate; TGA, thermogravimetric analysis; T_{MDR}, temperature of maximum degradation rate; VP, versatile peroxidase; XPS, X-ray photoelectron spectroscopy.

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commercial profitability. However, as the only aromatic polymer, the lignin fraction remains underutilized. Significant amounts of lignin are treated as a waste byproduct or burned on-site to generate heat and power [4]. Therefore, from the perspective of biorefinery profitability, lignin valorization is crucial for maximizing biomass utilization and decreasing waste generation [5].

Efficient fractionation of lignin from lignocellulosic biomass with favorable properties is a key step towards realizing lignin valorization. Traditional solvents (such as dilute acid, alkali solvents and organic solvents) are not suitable for lignin fractionation because they generate condensed lignin and/or pseudo-lignin [6]. Efforts devoted to lignin valorization have propelled the development of novel pretreatment methods using a new class of solvents, such as biomass-derived organic solvents (e.g., tetrahydrofuran and γ-valerolactone) [7] and ionic liquids [8,9]. However, some disadvantages, including high costs, complex synthesis procedures, high toxicity, or poor recoverability, may decrease commercial profitability and restrict their practical applications [10].

Recently, there has been great interest in using deep eutectic solvents (DESs) for fractionation due to their excellent capability of solubilizing lignin from lignocellulosic biomass [11–16]. DESs are liquid eutectic mixtures of a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD). Homogeneous liquid eutectic mixtures are formed by simply stirring two solid components in a certain ratio under mild conditions (low temperature and ambient pressure) for several minutes or hours. With this method, it is unnecessary to introduce additional solvents or complex purification steps. HBAs and HBDs are combined via hydrogen bond (H-bond) interactions. The formation of H-bonds is responsible for charge delocalization and a much lower melting point than that of the individual components. Since first introduced in 2004 [17], DESs have emerged as a promising new generation of green solvents. DESs fully agree with the 12 principles of green chemistry and have been used in many fields. DESs possess many favorable features, including the availability of raw materials, easy synthesis, tunable properties, biodegradability and satisfactory recyclability. These features also pave the way for their applications in lignin extraction and further valorization [18].

Renewable DESs are effective and eco-friendly solvents that can enable utilization of the lignin moiety in lignocellulosic biomass. DESs can be used in a myriad of applications as solvents, reaction media, or catalysts for lignin fractionation and valorization. The lignin fractionated by DESs can be applied in a wide range of application fields, from energy to various smart materials. Despite considerable efforts that have been devoted, a holistic review of DESs in lignin fractionation and valorization has yet to be explored. Several gaps in our fundamental knowledge and unsolved technical challenges remain undissolved. There are still issues with the type V DESs that needed to be addressed, such as the underlying mechanisms of intensification strategies with DESs treatment, a summary of DESs lignin's structure and properties, the mechanism of lignin dissolution in DESs and influencing factors, the aids of computational approaches for a deep understanding of the interactions between DESs and lignin at a molecular level, and the valorization of lignin in DESs medium.

In this review, a comprehensive summary of recent advances in the lignin fractionation and valorization using DESs is presented. Type V DES, the relationship between Kamlet–Taft solvatochromic parameters, and lignin fractionation and solubility are emphasized. Key factors that affect the lignin fractionation yield and lignin structure have been identified. In particular, recent progress is also highlighted in computation-based studies on the screening of DESs for lignin fractionation and the in-depth investigation of mechanisms. Moreover, the properties of DESs-fractionated lignin and available routes for its further upgrading routes are meticulously summarized. Finally, viewpoints on the research gaps and opportunities for the future development of lignin fractionation and upgrading are also given. This review provides insights into the fractionation and valorization of lignin by using tunable DESs, with the end goal of achieving a global low-carbon circular economy.

2. Lignin fractionation using deep eutectic solvents

2.1. Categories of DESs

The majority of the DESs reported to date are classified into five main categories based on the nature of the complexing agent, as shown in Fig. 1. Type I DESs are formed from quaternary ammonium salt (e.g., choline chloride, ChCl) and metal halide (e.g., $AlCl_3$), and has not been significantly applied to lignocellulose processing due to the high melting points of non-hydrated metal halides. Type II DESs mainly include a quaternary ammonium salt and a hydrated metal halide (e.g., $AlCl_3{\cdot}6H_2O$). These chemicals are more viable for industrial processes because they are inherently insensitive to air and moisture. Type III DESs mainly consist of a quaternary ammonium salt and various HBDs (e.g., carboxylic acids). This type of DES is most commonly used in the biomass processing, due to their low-cost starting materials and easy preparation. Type IV DESs include inorganic transition metals and HBDs (e.g., urea), even though metal salt does not normally ionize in non-aqueous media. Type V DESs are composed solely of non-ionic molecular substances and normally are hydrophobic [19]. For type V DESs, it is difficult to differentiate between HBAs and HBDs. An abnormally strong interaction stemming from the acidity difference of the phenolic and aliphatic hydroxyl (OH) groups was found to be the key to prepare type V DESs. Generally, type V DESs are hydrophobic and presents relatively higher volatility. Type V DESs possesses a significant advantage over the other four DES types because it allows for the potential recovery and regeneration of the eutectic solvent by evaporation. This characteristic provides more feasible possibilities for developing effective DES recycling technologies [19]. The type V DES was only recently identified, and thus, no study has been conducted to investigate the lignin fractionation efficiency using this DES. Given that type V DES exhibits highly efficient dissolution of lignin [20], further work is anticipated to extend type V DESs for lignin fractionation.

DESs can also be categorized based on other criteria, such as pH condition and miscibility with water. DESs are roughly categorized into three classes according to pH values, i.e., acidic, neutral, and basic DESs. The pH value of a certain DES varies depending on the nature of the starting materials, namely, functional groups existing in HBAs and

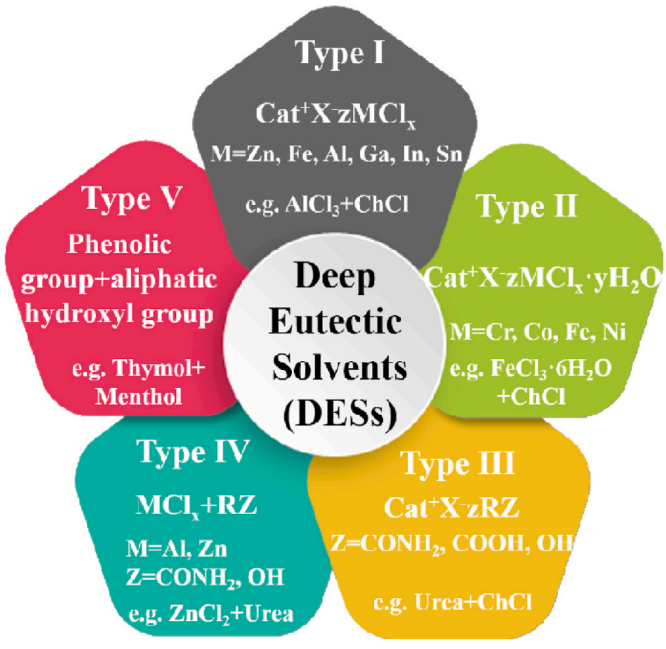


Fig. 1. General formula for the classification of DESs.

HBDs. There are also two types of DESs according to miscibility with water: hydrophilic and hydrophobic. A vast majority of DESs synthesized thus far are hydrophilic, and only a few DESs are hydrophobic. Hydrophobic DESs generally contain poorly water-soluble components, and can be further divided into two categories based on ionic compounds (e.g., tetraalkyl quaternary ammonium/phosphorous salts) and non-ionic compounds (e.g., monoterpenes) [21]. Hydrophobic DESs are an emerging subclass of DESs, and therefore, their capacity for lignin fractionation is presently an unexplored field, and further efforts are required.

2.2. Variation of DESs

Numerous chemicals are suitable for DES synthesis. Fig. 2 shows representative HBDs, HBAs, and additives for DESs synthesis. These constituents jointly determine the nature of the synthesized DESs and their behavior in lignin fractionation. A range of chemicals including acids, carbohydrates, poly-alcohols, amides, and phenolic compounds are mainly used as HBDs for lignin fractionation. According to the functional group of HBDs, DESs can be divided into four classifications. (1) Carboxyl acid-based DESs. Many substances with carboxyl groups can function as an HBD, such as monocarboxylic acids [e.g., formic acid (FA), acetic acid (AA), and lactic acid (LA)], dicarboxylic acids [e.g., oxalic acid (OA), malonic acid (MA), and glutaric acid (GA)], and tricarboxylic acids (e.g., citric acid). (2) Amine/amide-based DESs. This kind of DESs is composed of amine-based (e.g., mono-ethanolamine), amide-based (e.g., urea), or imidazole-based [e.g., 1,5-diazabicyclo (5.4.0)-5-undecene (DBU)] compounds, and usually exhibits alkaline behavior. (3) Polyalcohol/carbohydrate-based DESs. Many chemicals that contain hydroxyl groups can act as HBDs, such as polyalcohol [e.g., ethylene glycol (EG) and polyethylene glycol (PEG)] and carbohydrate-based DES (e.g., glucose), and usually exhibits neutral or near-neutral pH conditions. (4) Lignin-derived DESs. More recently, it has been reported that a series of phenolic compounds have been reported to form

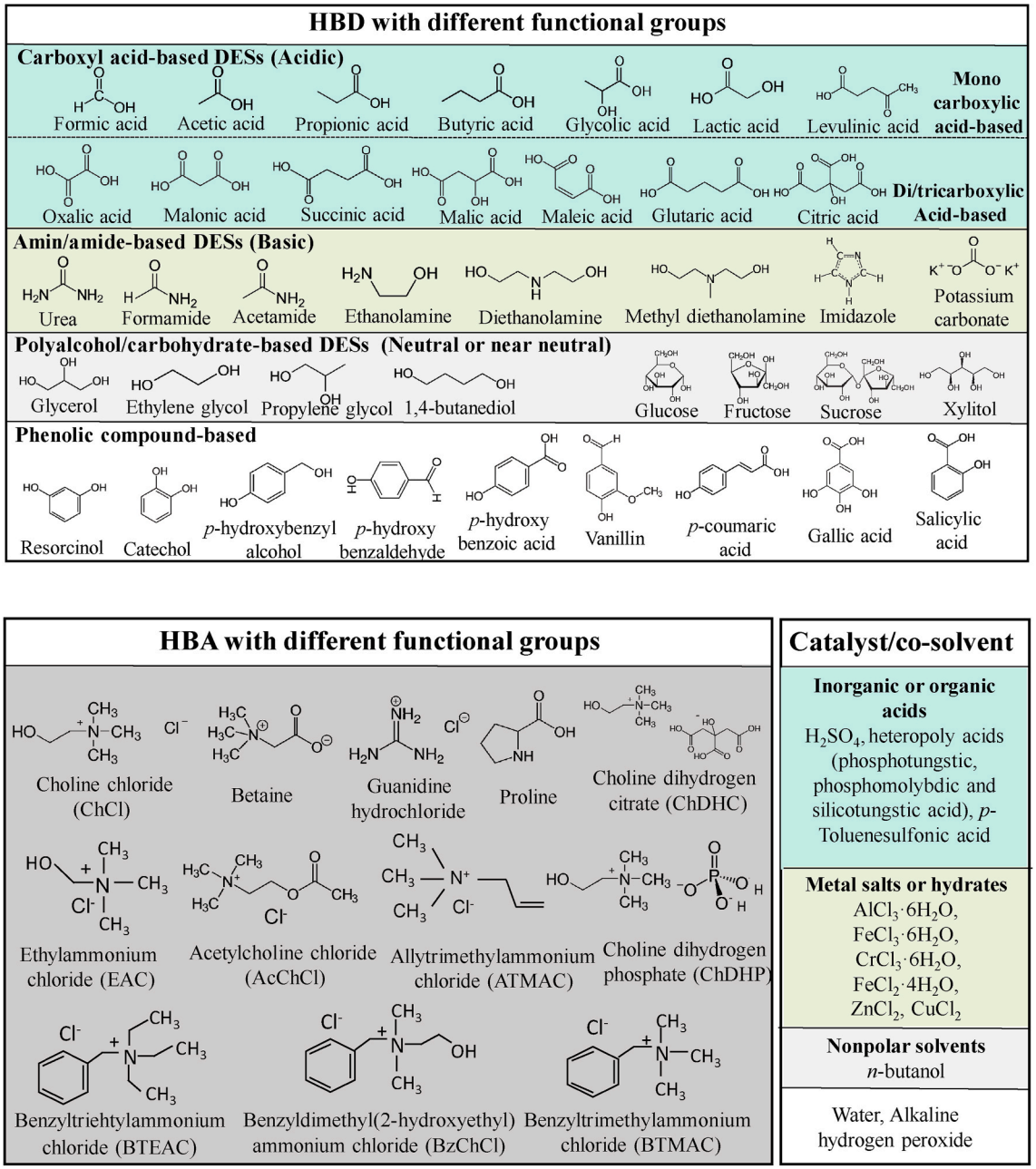


Fig. 2. Representative structures of HBD and HBA of DES as well as catalysts/co-solvents.

DESs with ChCl, such as vanillin, *p*-coumaric acid (PCA), catechol, *p*-hydroxybenzyl alcohol (PBA), *p*-hydroxybenzoic acid (PB), or *p*-hydroxybenzaldehyde (PHA).

HBA constituents play an imperative role in the performance of DESs, which affects the proton availability and reactivity. As shown in Fig. 2, a number of substances can function as HBAs. Among these HBAs, ChCl is the most adopted in lignin fractionation due to its benign properties, ability to rapidly form with various HBDs, and cost-effectiveness in lignin fractionation. In addition, several researchers have reported that certain HBDs can be used interchangeably as HBAs. Some chemicals, for example, LA, GH, and glucose, that consist of various functional groups (both electron-withdrawing and electron-donating), can function either as an HBD or HBA [22,23].

The synthesis of DESs is not limited to using only two constituents. Ternary DES can be constructed by mixing three compatible constituents. Generally, the third constituent is added as a catalyst or co-solvent to improve the DES fractionation performance. Some kinds of acids, metal halides, or organic solvents can be used as catalysts (listed in Fig. 2). The incorporation of these chemicals with HBAs and HBDs can tune the properties of DESs, and thus improve their lignin fractionation performance. For example, ethylene glycol is added into ChCl:OA to provide the desired stabilization function for efficient lignocellulose fractionation [24], neutral DESs (normally polyol-based, e.g., ChCl: glycerol) are intensified by incorporating a small amount of H_2SO_4 or *p*-toluenesulfonic acid [25–28].

2.3. Properties of DESs

2.3.1. Solvatochromic properties

The Kamlet–Taft (K–T) solvatochromic parameter is a quantitative tool reflecting the chemical properties of DESs. Three parameters, namely α, β, and π*, quantify hydrogen-bond- donating ability (acidity), hydrogen-bond-accepting ability (basicity), and di-polarity/ polarizability of the DESs, respectively. These parameters are determined by the nature of HBA and HBD constituents and their functionalities [4]. K–T solvatochromic parameters are able to interpret the relationship between DESs' properties and the efficiency of lignin fractionation, as well as the prediction of lignin solubility. DESs with higher α and β values exhibit stronger lignin fractionation ability [29]. The difference between the values of α and β is positively correlated with lignin solubility of DESs [30,31]. DESs with high α - β value exhibit high net H-bond-donating ability and form strong H-bonds with lignin, thus contributing to the degradation of lignin. Additionally, the π* values of DESs are also related to the solubilization and fractionation of lignin [20,32]. Apart from the K–T solvatochromic parameters, Hildebrand and Hansen solubility parameters are also used to predict the lignin solubility in various DESs with reasonable accuracy [33,34]. They are computed based on the predicted infinite dilution activity coefficient by molecular dynamic (MD) simulations, and can assist in the selection of suitable DESs for lignin fractionation.

Currently, a clear understanding regarding specific interaction between lignin and DESs governed by these parameters is lacking. Additional in-depth investigations are required in future works. Fortunately, DESs can be predictively tuned through changes in the variation of eutectic constituents based on some theories (for example, conductor-like screening models for real solvents, COSMO-RS). In consequence, with increased understanding of the structure-function relationship between solvatochromic parameters and lignin, a tunable DES pretreatment system can be rationally designed to fractionate lignin with high efficiencies and desirable properties for further valorization.

2.3.2. Physicochemical, electrochemical, and biological properties

The knowledge of physicochemical, electrochemical, and biological properties of DESs is also valuable to DESs practical applications. Key physicochemical properties of DESs include melting point (m_p), molar heat capacity (*Cp*), surface tension (γ), density (ρ), compressibility (k_s), speed of sound (u), viscosity (η), conductivity (σ), refractive index (n_D), and dielectric permittivity [35]. These properties depend on the liquid structure of DES. Additionally, water affects the structure and properties of DESs [36].

Vapor–liquid equilibria is informative and instructive for the developing effective recycle strategies for DESs. Generally, the higher the H-bond strength within a DES, the lower the melting point of the DES. However, the presence of extensive H-bonding networks within a certain DES may increase its viscosity. DESs viscosities are dependent on the nature of HBAs and HBDs, molar ratios, temperature, and the addition of co-solvent [37]. High DES viscosity is unfavorable for lignin fractionation due to the low rates of heat and mass transfer during DES pretreatment [23,37]. Additionally, because the surface tension and conductivity of DESs are closely related to viscosity, DESs with high viscosity normally possess high surface tension as well, with little conductivity [38]. Ultrasonication is a widely utilized strategy to intensify lignin fractionation [39]. Previous studies have shown that DESs with high viscosity and surface tension, and low diffusivity require a greater amount of ultrasonication force to guarantee the lignin fractionation efficiency [11,15,40].

Density is another important parameter of DESs. Density reveals the activity and molecular mobility of the solution. The density of DESs increases with the number of hydroxyl groups, and decreases with the increase of the alkyl chain length of HBDs. Generally, type I–IV DESs have higher densities than water (within the range 1.1–1.4 g cm^{-3}), whereas the density of type V DESs is lower than that of water [35]. From the view of industrial applications, the difference in the density between the DES and water should be sufficiently large. The density of DES should be fine-tuned through designing structures to suit the lignin fractionation.

The acidity/basicity of a DES is one reflection of its liquid structure and it varies according to the nature and ratio of their counterparts. It can be quantified by the apparent pH, bromophenol blue indicator, and Hammett acidity function [35]. The acidity and basicity of DESs play important roles in the lignin fractionation. According to the acidity and basicity of DESs, the transformations of lignin during fractionation by DESs are classified into three pathways, namely acidic, alkaline, and neutral [6]. Based on the design of hydrogen bond acidity and alkalinity, the acidity and basicity of DESs can be adjusted to match the lignin fractionation and its further valorization.

Electrical conductivity indicates the strength with which DESs can conduct electricity. It is affected by the number of charged substances and their mobilities. DESs normally exhibit poor conductivity (lower than 2 mS cm^{-1}) [35]. To boost lignin fractionation, microwave irradiation is generally applied [41]. When using microwave heating, DESs with high electrical conductivity and dielectric constant exhibit effective delignification of lignocellulosic biomass through molecular collisions [42,43]. Additionally, sufficiently high conductivity is essential to perform the electrochemical transformation of lignin in DES systems [44,45]. The electrical conductivity of DESs can be customized by changing the components and their molar ratio. In the following research, the development of novel DESs can be broadly implemented in a lignin biorefinery.

The biological properties of DESs normally refer to low or non-toxicity, and high biodegradability and biocompatibility. DESs are compatible with many microbes and enzymes, and are used as reaction media to produce microbial lipids [46] and biogas [47]. DESs are generally considered as green, especially for natural DESs which are made up of naturally occurring components. Natural DESs are regarded as GRAS (Generally Recognized as Safe), and are eligible to be used in the food and pharmaceutical industries [48]. Although biological properties have little impact on lignin fractionation yield and structure, they are important characteristics of DESs because they affect the techno-economic evaluation of this strategy, life-cycle management, and practical application, thereby influencing holistic analysis.

2.4. Effect of DESs on lignin fractionation

The initial and most crucial step during lignin valorization is the efficient fractionation of lignin with favorable properties. Type III DES is the most frequently studied category for lignin fractionation from lignocellulosic biomass. There has been only one report thus far where a type II DES (ChCl:$CuCl_2{\cdot}2H_2O$) was used to fractionate lignin from oil palm leaves, and the lignin fractionation yield reached 55.14% [49]. Aside from the botanical source and treatment conditions, the lignin fractionation performance of DESs is dependent on the DESs constituents and molar ratios [50]. It has been reported that 42 parameters influence the delignification efficiency of DESs [51]. Table 1 shows the DES pretreatment conditions and the corresponding performance of lignin fractionation from selected reported works using various DESs. The lignin fractionation performance of DESs will be discussed in the following section based on their constituents.

2.4.1. Functional group type and quantity in HBD

Considerable amounts of chemicals are suitable as HBDs for DES synthesis. Depending on the nature and number of functional groups in HBDs, the lignin fractionation yield varies accordingly. Consequently, unveiling the correlation between functional groups in HBDs and lignin extraction is important to tune DESs for the fractionation of lignin with tailored properties.

2.4.1.1. Carboxyl groups in HBDs. A variety of chemicals containing carboxyl groups have been used as HBDs to synthesize DESs. Carboxyl acid-based DESs are the most widely used DES for lignin fractionation due to their high efficiency. As seen in Table 1, ChCl:LA (1:10) is capable of removing approximately 90% of the lignin from various types of lignocellulose such as corncob [52], willow [53], *Eucalyptus* [54], and *Cunninghamia lanceolate* [55]. This effective lignin fractionation performance mainly originates from the presence of active protons supplied by carboxylic groups. These active protons promote proton-catalyzed cleavage of various chemical linkages (e.g., ether and ester bonds) in lignin-carbohydrate complex (LCC) and lignin. Generally, carboxylic acid-based DESs exhibit a stronger performance in lignin fractionation than those containing other groups regardless of the HBAs used.

A comprehensive understanding of the correlation between the lignin fractionation performance and the structure-function of DESs is essential to guide the selection of desirable DESs conducive for further lignin valorization. The effective delignification of carboxyl acid-based DESs is closely related to the acid strength, amount of carboxyl groups, and the molar ratio. Fig. 3a shows that the lignin fractionation yield increases with acid strength of monocarboxylic acid-based DESs [50]. Overall, HBDs containing shorter alkyl chains are more efficient in lignin fractionation than those with longer chains. This is probably the case because of the electron donation from the alkyl group to the oxygen in the OH group, thus intensifying the H-bonding strength. For dicarboxylic acids, the lignin fractionation exhibited a negative relationship with the amount or strength of the acid. The monocarboxylic acid-based HBDs possess the optimal lignin fractionation capacity compared with the di- and tricarboxylic acids [56].

When the HBD contains more than one carboxyl group, efficient interactions between the resulted DES and lignin are impaired. Two possible reasons might explain the weak delignification of di- and tricarboxylic acids: (1) extra carboxyl groups in HBDs formed more H-bonds and extensive dimer chains, thus reducing the mass transfer rate of the solvent system; (2) the carboxyl groups in the HBD may easily be activated, and thus, release carbon dioxide at relatively high temperatures. In addition, the molar ratio of HBAs and HBDs also affects lignin fractionation performance. By increasing the molar ratio of LA to ChCl from 2 to 15, the delignification of corncob increased from 64.7% to 93.1% [52]. Similar findings were also reported when using ChCl:LA fractionate lignin from oil palm empty fruit bunches [50] and *Cunninghamia lanceolata* [55]. Such an increase in delignification could be attributed to more active protons donated by LA, which facilitated proton-catalyzed cleavage of various bonds in LCC.

2.4.1.2. Hydroxyl groups in HBDs. Various chemicals with hydroxyl groups are also widely used as HBDs to synthesize DESs [57]. When paired with a non-toxic HBA, the resulted DESs usually exhibit near neutral pH and show good biocompatibility with enzymes [35] and microorganisms [46,47]. Nevertheless, poor results are obtained for lignin dissolution and fractionation performed by most DESs of this type, even under microwave-assisted [42] or severe pretreatment conditions [58]. However, 71.3% and 87.6% of lignin were removed from corncob using ChCl:glycerol and ChCl:EG under facile pretreatment conditions [52]. The vast difference in these results could be the varying recalcitrance degree in different biomass types or reaction conditions employed (Table 1).

The number and position of the hydroxyl group have significant impacts on the lignin fractionation yield from lignocellulosic biomass. Lignin fractionation yield is inversely correlated with the number of hydroxyl groups in the HBD. As shown in Fig. 3b, DESs that are composed of xylitol (with five hydroxyl groups) produced the lowest lignin yield, followed by DESs that are composed of glycerol (with three hydroxyl groups), as well as DESs that are composed of EG (with two hydroxyl groups), regardless of the HBAs [22]. The extra hydroxyl groups in HBDs form a more extensive H-bond network, resulting in higher viscosity of DESs. Hence, additional energy is required to loosen the bonded structures to initiate interactions with lignin. Moreover, the presence of an extensive H-bond network weakens the ability of the HBA to compete with intra-molecular bonding in the lignin moieties of biomass [29]. Furthermore, despite bearing the same number of hydroxyl groups, the position of the hydroxyl groups slightly affects the lignin fractionation yield. For example, ChCl:1,3-propanediol fractionated more lignin from rice straw than ChCl:1,2-propanediol. This was possibly ascribed to the stronger hydrophobicity of 1,3-propanediol--based DESs, which promoted lignin fractionation using the hydrophobic phenylpropanoid skeleton [22]. The presence of hydroxyl groups in an α-hydroxyl acid-based DES conferred a positive effect on lignin fractionation. This was possibly because that the presence of hydroxyl groups in α-hydroxyl acid increased the polarity of the DES. Hence, the increased polarity resulted in stronger interactions between the hydroxyl group in the HBDs and the free/etherified hydroxyl groups in lignin [50]. However, these explanations are speculative and remain to be validated. More in-depth investigation on the synergistic effects between the different functional groups and their effects on lignin fractionation is imperative.

2.4.1.3. Amine/amide groups in HBDs. As seen in Table 1, chemicals with amine/amide groups are also used as HBDs for preparing DESs to fractionate lignin. The effect of amine/amide groups in HBDs on the performance of lignin fractionation is less investigated as compared to carboxylic acid-based and polyalcohol-based DESs. This kind of DESs normally are basic with moderate lignin fractionation performance [23]. The presence of amine/amide groups in HBDs endows the DES with strong basicity. This facilitates lignin fractionation by loosening various the chemical bonds of LCC and contributes to selective dissolution through the deprotonation of phenolic moieties in lignin. With the decrease in basicity (reflected in the pH values) from ChCl:monoethanolamine (pH value: 10.9), ChCl:diethanolamine (pH value: 10.5), and ChCl:methyldiethanolamine (pH value: 10.4) to ChCl:urea (pH value: 8.2), the lignin removal rate significantly decreased from 81.0%, 73.5%, and 44.6%, to 27.7%, respectively [59]. This is mainly ascribed to the fact that the phenolic OH moieties of lignin are deprotonated in alkaline solutions. The stronger the basicity of DES, the greater the number of moieties that are deprotonated. In addition, the number of amine groups in the HBDs affects the lignin fractionation yield. As

Table 1
Performance of various DESs on lignin fractionation.

DESs (HBA:HBD)	Biomass	Conditions: Temperature (°C), Time (h) and Solid loading (wt %)	Lignin fractionation yield/removal rate/ delignification (%)	References
Acidic DESs Carboxyl-containing				
ChCl:LA (1:2)	Douglas fir	145 °C, 6 h, 10%	58.2	[32]
	Poplar		78.5	
	Sorghum		78	[93]
	Walnut endocarps cells		64.3	[95]
	Peach endocarp cells		70.2	
	Corncob	90 °C, 24 h, 5%	64.7	[52]
	Poplar	110 °C, 1.5 h, 10%	31.1	[101]
		130 °C, 1.5 h, 10%	66.4	
	Wheat straw	120 °C, 12 h, 10%	78.6	[94]
	Corn straw	120 °C, 6 h, 5%	51.84	[89]
	Castor seed coats	160 °C, 8 h, 10%	34	[58]
ChCl:LA (1:5)	Rice straw	60 °C, 12 h, 5%	60	[69]
	Corncob	90 °C, 24 h, 5%	77.9	[52]
	Oil palm empty fruit bunch	120 °C, 8 h, 10%	88	[23]
ChCl:LA (1:9)	Poplar	120 °C, 6 h	95	[56]
ChCl:LA (1:10)	Corncob	90 °C, 24 h, 5%	86.1	[52]
	Poplar	110 °C, 1.5 h, 10%	51.5	[101]
		130 °C, 1.5 h, 10%	89.3	
	Willow	120 °C, 12 h	91.8	[53]
	Eucalyptus camaldulensis	130 °C, 6 h, 10%	93.2	[54]
	Cunninghamia lanceolata	120 °C, 12 h, 3.3%	81.73	[55]
ChCl:LA (1:15)	Corncob	90 °C, 24 h, 5%	93.1	[52]
	Oil palm empty fruit bunch	120 °C, 8 h, 10%	61	[50]
ChCl:AA (1:2)	Castor seed coats	160 °C, 8 h, 10%	31	[58]
ChCl:OA (1:1)	Corncob	90 °C, 24 h, 5%	98.5	[52]
ChCl:FA (1:2)	Oil palm empty fruit bunch	120 °C, 8 h, 10%	50	[87]
	Castor seed coats	160 °C, 8 h, 10%	33	[58]
ChCl:MA (1:1)	Corncob	90 °C, 24 h, 5%	56.5	[52]
	Rice straw	80 °C, 6 h, 5%	51.3	[22]
	Castor seed coats	160 °C, 8 h, 10%	3	[58]
ChCl:glycolic acid (1:2)	Corncob	90 °C, 24 h, 5%	56.4	[52]
ChCl:levulinic acid (1:2)			43.0	
ChCl:malic acid (1:1)			22.4	
ChCl:glutaric acid (1:1)			34.3	
ChCl:citric acid (1:1)	Oil palm empty fruit bunch	120 °C, 8 h, 10%	20.6	[50]
ChCl:propionic acid (1:2)			20.4	
ChCl:succinic acid (1:1)			10.7	
betaine:LA (1:2)	Rice straw	60 °C, 12 h, 5%	52	[69]
	Corn straw	120 °C, 6 h, 5%	35.86	[89]
BTMAC:FA (1:2)	Pine	150 °C, 2 h, 5%	80.60	[83]
BTMAC:LA (1:2)	Corncob	140 °C, 2 h, 5%	63.4	[70]
BTEAC:LA (1:2)			56.6	
TEBAC:LA (1:9)	Wheat straw	100 °C, 10 h, 1:15	79.73	[86]
proline:malic acid (3:1)	Rice straw	120 °C, 12 h, 5%	48.7	[73]
Neutral DESs				
ChCl:glycerol (1:2)	Corncob	90 °C, 24 h, 5%	71.3	[52]
	Switchgrass	110 °C, 1 h, 10%	76.6	[26]
	Wheat straw	90 °C, 12 h, 5%	24.7	[59]
	Castor seed coats	160 °C, 8 h, 10%	4	[58]
ChCl:EG (1:1)	Rice straw	120 °C, 6 h, 5%	26.16	[22]
ChCl:EG (1:2)	Corncob	90 °C, 24 h, 5%	87.6	[52]
	Castor seed coats	160 °C, 8 h, 10%	8	[58]
ChCl:1,2-propanediol (1:1)	Rice straw	80 °C, 6 h, 5%	26.5	[22]
ChCl:1,3-propanediol (1:1)			29.2	
ChCl:glucose (1:1)	Oil palm empty fruit bunch	120 °C, 8 h, 10%	17	[23]
Basic DESs				
ChCl:urea (1:2)	Oil palm empty fruit bunch	120 °C, 8 h, 10%	34	[23]
	Wheat straw	90 °C, 12 h, 5%	27.7	[59]
ChCl:formamide (1:1)	Rice straw	120 °C, 6 h, 5%	45.8	[22]
ChCl:monoethanolamine (1:6)	Wheat straw	90 °C, 12 h, 5%	81.0	[59]
ChCl:diethanolamine (1:8)			73.5	
ChCl:methyldiethanolamine (1:10)			44.6	
ChCl:acetamide (1:2)			3.4	
K_2CO_3:glycerol (1:6)	Oil palm empty fruit bunch	120 °C, 8 h, 10%	51	[23]

(*continued on next page*)

Table 1 (*continued*)

DESs (HBA:HBD)	Biomass	Conditions: Temperature (°C), Time (h) and Solid loading (wt %)	Lignin fractionation yield/removal rate/ delignification (%)	References
Lignin-derived compounds-based DESs hydrophobic				
ChCl:PCA (1:1)	Switchgrass	160 °C, 3 h, 5%	60.8	[61]
	Poplar	160 °C, 3 h, 10%	15.5	[63]
	Cortex albiziae	160 °C, 7 h, 10%	30.79	[62]
		180 °C, 7 h, 10%	64.03	
ChCl:PB (3:2)	Poplar	160 °C, 3 h, 10%	69.0	[63]
ChCl:vanillin (1:2)	Switchgrass	160 °C, 3 h, 5%	52.5	[61]
	Arabidopsis thaliana mutant	80 °C, 3 h, 5%	24.6	[64]
ChCl:catechol (1:1)	Switchgrass	160 °C, 3 h, 5%	49.0	[61]
ChCl:PBA (1:1)			0.4	
ChCl:PHA (1:2)	*Arabidopsis thaliana* mutant	80 °C, 3 h, 5%	32.4	[64]
	Poplar	160 °C, 3 h, 10%	30.7	[63]
ChCl:GA (2:1)	Poplar	140 °C, 3 h, 5%	50	[65]
Ternary DESs				
ChCl:glycerol:LA	Wheat straw	120 °C, 0.5 h, 2%	45	[78]
ChCl:tartaric acid:LA (1:1:4)	*Pinus pinaster* Ait.	150 °C, 2 h, 10%	95	[37]
ChCl:glycerol:$AlCl_3{\cdot}6H_2O$ (1:2:0.28)	Poplar	120 °C, 4 h, 5%	95.46	[29]
ChCl:EG + H_2SO_4+water (1:2 + 1% + 10%)	Switchgrass	130 °C, 0.5 h, 20%	57.63	[25]
		130 °C, 0.5 h, 27%	51.77	
ChCl:glycerol + H_2SO_4 (1:2 + 0.9%)	Switchgrass	120 °C, 1 h, 10%	63.84	[26]
Propionic acid:Urea + PTSA (1:2 + 25%)	*Eucalyptus globulus*	190 °C, 8 h, 5%	80.64	[28]
ChCl:EG + H_2SO_4+water (1:2 + 1% + 20%)	Switchgrass	130 °C, 0.5 h, 10%	80	[27]
ChCl:$CuCl_2{\cdot}2H_2O$	Oil palm fronds	90 °C, 3 h, 10%, combined with alkaline hydrogen peroxide	55.4	[49]

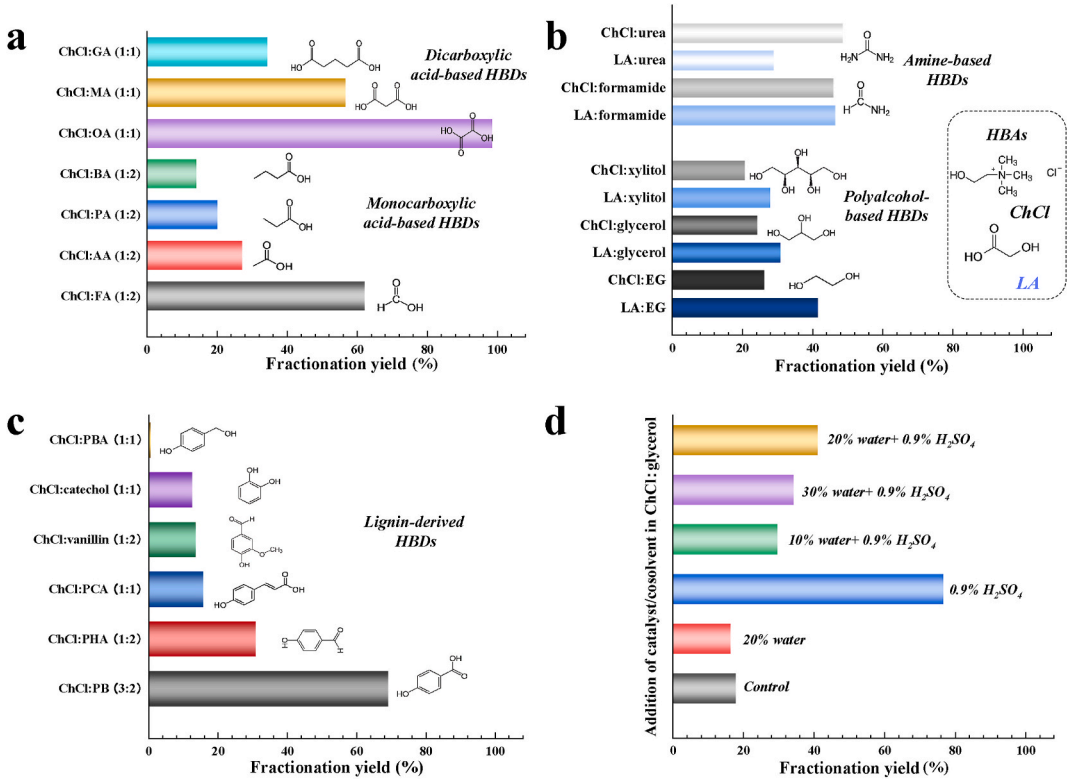


Fig. 3. Effect of DESs on lignin fractionation yield: (a) number of function groups and alkyl chain length in HBDs [50], (b) type of functional group in HBA [22], (c) lignin-derived HBDs [61,63], and (d) addition of catalyst/co-solvent in ChCl:glycerol (1:2) [26].

depicted in Fig. 3b, ChCl:urea achieved a higher (48.52%) lignin yield from rice straw than ChCl:formamide (45.83%) [22]. Currently, there have been few studies that have reported tuning this kind of DESs for improved lignin fractionation yield. Therefore, greater efforts are required to better understand their delignification mechanism and their influence on the properties of the extracted lignin.

2.4.1.4. Lignin-derived HBDs. DESs containing phenyl groups are supposed to exhibit great potential for dissolving lignin from lignocellulosic biomass. This is attributed to the ability of phenolic compounds to form

new strong π–π stacking interactions and H-bonds with lignin moieties, thus reducing or destroying the original interactions in LCC in lignocellulosic biomass [60]. Recently, a number of lignin-derived compounds, such as vanillin [61], catechol, PCA [62], PB [63], PHA [64], PBA, gallic acid [65], and guaiacol [66], have been reported as HBDs to synthesize DESs for biomass pretreatment.

The efficiency of lignin fractionation closely depends on the structure of lignin-derived compounds. Ragauskas' research group was the first to attempt to use lignin-derived DESs to fractionate lignin [61]. As shown in Fig. 3c, ChCl:PCA achieved the highest lignin removal rate, followed by ChCl:vanillin, ChCl:catechol, and ChCl:PBA. In another study, they found that the delignification performance of poplar using three lignin-derived DESs was in the following order: ChCl:PB (69.0%) > ChCl:PHA (30.7%) > ChCl:PCA (15.5%). The high delignification efficiency of PB was attributed to the presence of more active protons donated by carboxyl and phenolic OH groups as well as phenyl groups in PB. These active protons facilitated proton-induced bond cleavage in lignin [63]. Subsequently, this research group achieved a closed-loop biorefinery using ChCl:vanillin and ChCl:PHA to pretreat transgenic *Arabidopsis thaliana* [64]. In this route, 64.1% lignin was removed, and the produced lignin could be further processed to supply more phenolic compounds for DESs synthesis. Following this pioneering work, the performances of lignin fractionation using lignin-derived DESs have been validated from *Cortex albiziae* herb residues [62], woody biomass [63], hybrid poplar [65], and wheat straw [66].

Lignin-derived DESs hold great promise in lignin fractionation from lignocellulosic biomass in a one-pot procedure. In addition, the renewable DESs prepared from lignin-based components can aid the development of a closed-loop biorefinery, which further assists in achieving a sustainable bio-economy. Because a variety of lignin-derived compounds are available, further screening of such compounds will contribute to the development of additional desirable DESs for the fractionation and valorization of lignin.

2.4.2. Functional group type in HBAs

HBAs also play a significant role in the performance of lignin fractionation. DESs display more optimal lignin fractionation than the corresponding acids alone. For example, the lignin yield of pure LA was 41.52%, whereas the lignin yield of ChCl:LA (1:10) reached approximately 70% [53]. A recent study determined that the effective delignification performance of ChCl-based DESs stems from the highly electronegative chlorine ion, which tends to form H-bonds with the hydroxyl groups in lignin [67]. More recent investigations have found that the halide anions in the HBA contributes to the breakage of β–O–4' bonds and prevents lignin condensation [27,68].

The efficiency of lignin removal is also closely affected by the coordinating ability of cations [69], functional groups [57] and the length of the alkyl groups [70] in the HBA constituents. With respect to the hydroxyl-containing HBDs, the lignin fractionation yields of DESs that used LA as HBAs were all higher than those of corresponding DESs that used ChCl as HBAs (as illustrated in Fig. 3b) [22]. In the pretreatment of corncob, there was a relatively higher lignin removal rate for BTMAC:LA as compared with BTEAC:LA [70]. This may be attributed to the shorter alkyl chain for BTMAC that is less bulky than that for BTEAC, and hence, less steric hindrance occurs during lignin removal. Collectively, a suitable selection of HBAs largely determines the DES capacity of lignin fractionation. Hence, identifying the optimal combinations of DESs is the first step in lignin biorefinery.

2.4.3. Intensifying DESs with catalysts/additives

To improve the DES fractionation performance under facile conditions, an additional constituent is usually added (Fig. 2), which functions as a catalyst or co-solvent. Various acids, solvents and water are capable of increasing the lignin fractionation yield (Table 1).

Incorporation of small amounts of H_2SO_4 (as shown in Fig. 3d) into ChCl:glycerol significantly improved the lignin fractionation performances from switchgrass [26]. The addition of Lewis acids greatly increased the delignification of *Pennisetum* energy grass due to the enhanced H-bond acceptor strength and acidity in ChCl:glycerol [71]. Particularly, the coordination of $AlCl_3{\cdot}6H_2O$ into ChCl:glycerol markedly enhanced the lignin fractionation efficiency from 3.61% to 95.46% [29]. Significant improved efficiencies of delignification of vegetable wastes were also achieved by the addition of $AlCl_3{\cdot}6H_2O$ into ChCl:glycerol [11]. The introduction of $AlCl_3{\cdot}6H_2O$ provides active protons and acidic sites that facilitate proton-catalyzed bond cleavage in lignin. Additionally, three heteropoly acids (phosphotungstic, phosphomolybdic, and silicotungstic acids) greatly improved the delignification performance of neutral DESs [72]. Aside from the above-mentioned inorganic acids, organic acids have also been applied. The integration of *p*-toluenesulfonic acid with propionic acid:urea resulted in the detachment of 80.64% of lignin to detach from *Eucalyptus globulus* at 89 °C for 8 h [28]. Except acids, some organic solvents that highly dissolve lignin are conducive for boosting lignin fractionation. The incorporation of ethylene glycol into ChCl:OA significantly increased the yield of lignin fractionation [24].

The addition of water to a small amount of ChCl:LA [69] or ChCl:OA [73] was favorable to increase the effectiveness of lignin fractionation. The addition of water probably reduces the viscosity of DESs and improves diffusivity, resulting in increased mass transfer and interactions between the DES and lignin. However, an opposite trend has also been reported. As shown in Fig. 3d, the addition of water at 20–40 wt% had a negative effect on the lignin fractionation of acidified ChCl:glycerol. This may have occurred because that water competes for association with the chloride anions [74]. When water is added into a DES, it can disrupt the original H-bond network and a new H-bond network can form. The change in the H-bond network alters the supramolecular network of DES. Most relevant eutectic features are then lost. The H-bond network may be intensified or weakened, depending on the nature of the DES constituents. Consequently, the addition of water may be beneficial or harmful for the lignin fractionation. In addition, the removal of excess water to obtain a pure DES requires more energy during recycling. Whether water is added or not, the addition amount of water should be carefully managed to maximize lignin fractionation.

2.5. Integration of DES fractionation with other technologies

Several biomass intensification technologies (such as ultrasound, microwave, hydrothermal, and torrefaction) have been adopted to boost the performance of lignin fractionation by increasing its accessibility to DESs. In this section, the impact of the combined strategies, mechanisms and their performances in lignin fractionation are discussed.

2.5.1. Ultrasonication-assisted DES strategy

Ultrasound irradiations are known to intensify the lignin fractionation from lignocellulosic biomass through cavitation and mechanical effects. Moreover, ultrasonic energy provides a unique physicochemical environment that disrupts the lignin barrier. As shown in Fig. 4, the cavitation of micro-bubbles, microjets shooting on the surface, and strong agitation that results due to mechano-acoustic effects during ultrasonication increase the micro-pores for higher surface area contact with DESs. Consequently, ultrasonication enhances the fractionation of lignin. Significant synergistic effects between ultrasound and various DESs have been observed in the fractionation of lignin from corn stalk [75], *Parthenium hysterophorus* [76], and oil palm fronds [77]. In our previous studies, efficient ultrasound-assisted DES pretreatments have been developed to deconstruct the recalcitrant structure of sugarcane bagasse [12,16] and watermelon rind [13,15] for lignocellulose fractionation.

Among various ultrasonication parameters, power, frequency, and time significantly affect delignification performance [15]. More recently, frequency mode (i.e., sweeping frequency or fixed frequency) has been identified as an important influential factor on lignin

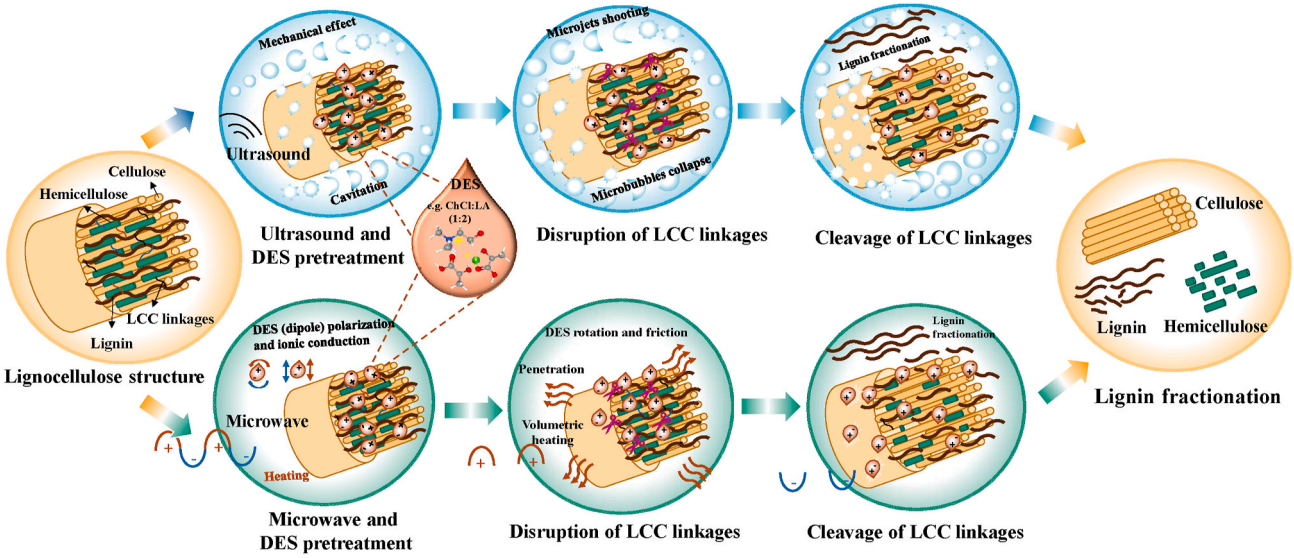


Fig. 4. Working mechanisms of ultrasound and microwave assisted DES pretreatment.

fractionation. There was a relatively higher synergism with DESs when sweeping frequency ultrasound was used for lignin fractionation rather than fixed frequency ultrasound [16]. In summary, ultrasound intensity (as well as power densit and frequency modes) and duration should be carefully designated to maximize the delignification efficacy and minimize costs. Moreover, future works are needed to study the utilization of different ultrasonication media to enhance lignin fractionation efficiency.

2.5.2. *Microwave-assisted DES strategy*

Microwave technology has been proved to be a promising strategy for biomass pretreatment. It can reduce process time, energy consumption, and the severity of DES pretreatments [78]. Microwaves interact with polar molecules, which leads to rapid, uniform volumetric heating through dipole rotation and ionic conduction with a lower energy loss [11]. Because DESs are electric dipoles, microwave irradiation can maximize the ionic character of DESs and increase their molecular polarity. As shown in Fig. 4, the movement of polar molecules in DESs may enhance the penetration of DES solvent into biomass pores and provide energy to activate the lignin bond-breaking required for lignin solvation.

With a microwave-assisted DES strategy, less process time and energy consumption is required for lignin fractionation from various biomass sources, such as vegetable wastes [11], corn straw [14], poplar [60], *miscanthus* and birchwood [42], wheat straw [79], and rice hulls [80]. Microwave heating not only enhances lignin fractionation through faster heating rates but it also promotes selective bond cleavages during lignin depolymerization. Certain lignin bonds including C_β–O in the β–O–4 linkage are stretched under the electric field imparted during microwave irradiation [43]. A microwave-assisted DES strategy offers intriguing future perspectives for lignin fractionation. However, many factors, such as microwave penetration depth, DESs properties, type of biomass, and process-related costs must be taken into consideration in the practical production. In the future works, these technical and economic issues should be solved to drive the fractionation of lignin in a more cost-efficient manner.

2.5.3. *Integrated strategy of DES coupled with other pretreatments*

DESs have been sequentially integrated with other pretreatment strategies, such as inorganic salt, hydrothermal, and torrefaction pretreatments, to enhance the lignin fractionation efficiency. A two-stage sequential pretreatment with dilute sodium hydroxide before or after with DES (ChCl:urea) treatment led to higher lignin fractionation (79.3% and 78.4%) from sorghum straw, when compared with pretreatment of DES alone (49.0%) [55]. The hydrothermal pretreatment prior to DES pretreatment disrupted biomass cell walls and cleaved lignin into small fragments. It significantly reduced the biomass recalcitrance and boosted the efficiency of lignin fractionation using DESs [81]. These reports demonstrated that the integrated strategy of DES coupled with other pretreatments can be tailored to efficiently fractionate lignin as well as cellulose and hemicellulose. This intensified approach holds great promise in the complete utilization of lignocellulosic biomass and will pave the way for a closed-loop biorefinery process.

2.6. *Recycling and reusability of DESs*

Despite the potential shown by DESs in lignin fractionation from lignocellulosic biomass, one significant issue is the recovery and recycling of DESs. This is an essential prerequisite for commercial applications considering economically feasible and sustainable bio-refineries. Recycling reusable DESs could reduce chemical consumption, minimize waste stream generation, and lower the overall cost. DESs are commonly recovered and recycled by evaporating the anti-solvent (such as water, ethanol, ethanol/water, acetone/water) coupled or uncoupled with removal of impurities, followed by (ultra)filtration of the liquid phase. In addition to the evaporation, electrodialysis has shown high effectiveness in the recovery of DES constituents [82].

The reusability of DESs can be evaluated by recovery yield [61], and physiochemical and fundamental structural properties of the recycled DES [62], as well as enzymatic hydrolysis [12] and delignification performances [83]. A lifetime and recyclability experiment with a DES showed that the recovery yield of the DES was at least 90%, and the fundamental structural properties of the recycled DES were almost unchanged. Concurrently, the efficiency of lignin fractionation was largely maintained throughout the recycling periods [54]. Various DESs have been successfully recycled via evaporation for four or five pretreatment cycles with acceptable delignification efficacies. However, recycling DESs by evaporation is often energy-intensive, and in the future, a low-energy approach for recycling should be developed.

The recycling of DESs is limited by their thermal instability and susceptibility to contaminants [84]. Concerning their thermal instability, the release of trimethylamine from ChCl-based DESs or the release of amine from glycine betaine-based DESs at elevated temperatures (i.e., Hoffman elimination reaction) is a detrimental component for its industrial viability. Additionally, the recovered DES solution may contain

a range of solutes, such as various monomer sugars, furfural, carboxylic acids, and phenolic compounds. These impurities may modify the properties of the DES, and thus affect its fractionation effectiveness. The slight decrease in the enzymatic digestibility and property changes of the isolated lignin that occur with recycled DESs stem from the impurities in the DESs after each recycling. However, no study has been conducted to investigate the removal of impurities in the recycled DES solutions. These limitations must be overcome, and effective DES recycling technologies should be developed before DESs can be broadly consolidated on an industrial scale for lignin fractionation and valorization.

2.7. Transformation of lignin during DES fractionation

Elucidating the structural changes of lignin during DES pretreatment is not only essential to facilitate the optimization of the fractionation process but it is also considerably important for lignin valorization. Various state-of-the-art analysis techniques such as quantitative ^{13}C nuclear magnetic resonance (NMR), ^{31}P NMR, two-dimensional heteronuclear single quantum correlation (2D HSQC), gel permeation chromatography (GPC), size exclusion chromatography (SEC), pyrolysis-gas chromatography-mass spectrometry (Py-GC/MS), and attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy can provide insights into the structural changes of lignin [85]. Due to the superior performance of DESs in lignin fractionation, the mechanism regarding the interaction of DESs with lignin is being actively studied. Hong et al. have completely and comprehensively reviewed the structural transformation during the lignin fractionation using DESs [6]. The proposed lignin reaction pathways during DES pretreatment include dehydration, depolymerization, repolymerization, acylation, and demethoxylation, dehydration. However, the exact mechanism for bond cleavage in DESs is complicated and requires additional in-depth investigation.

3. Properties of DES lignin

The properties of the DESs-extracted lignin significantly impact its further valorization. Fig. 5 shows that the properties of lignin include purity, molecular structure, thermal characteristics, and other properties such as biological features. These properties can be properly analyzed and characterized by classical analytical methods, such as the Klason method, derivatization followed by reductive cleavage (DFRC), and alkaline nitrobenzene oxidation (NBO), together with various instrumental tools such as HSQC, NMR, GPC, X-ray photoelectron spectroscopy (XPS), thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), ATR-FTIR, and Py-GC/MS. Although only a few studies have systematically investigated DES lignin, some desirable properties have been revealed (Table 2). These appealing properties imply good valorization potential for this new source of technical lignin.

3.1. Purity

The purity of DES lignin is usually calculated based on the classical Klason method. Lignin fractionated by DES is superior to that produced by conventional procedures in terms of purity. The purity of DES lignin available is normally in the range of 75.69–98% (Table 2), which is advantageous for further valorization. For example, the purity of lignin extracted by ChCl:LA from poplar is 98% [56]. DES lignin is generally recovered or regenerated by precipitation and purification via adding anti-solvents. The impurities in DES lignin are normally DESs and polysaccharide residues. Because DESs have high selectivity for lignin, the purity of lignin is insignificantly influenced by DES configuration. For example, the purities of all DES lignin are greater than 85% using the tested ChCl-based DESs with various HBDs (Fig. 6a) [58]. Nonetheless,



Fig. 5. Representative instrumental tools used to characterize various properties of lignin.

Table 2
Properties of linin fractionated by various DESs.

DES and biomass type	Reaction conditions	Precipitation and purification solvents	Purity	Structure			Reference
				M_W and DPI	NMR	Thermal	
ChCl:LA Douglas fir	145 °C, 6 h, 10%	Water; Ethanol/water mixture (1:9, v/v)	95.4	1340;/	Negligible amount of ether linkages, high phenolic OH group, absence of the β–O–4 ether linkage, appreciable amount of Hibbert's ketone, less condensed	/	[32]
ChCl:LA (1:2) Sorghum	145 °C, 6 h, 10%	Water/ethanol (1:2); Ethanol/water mixture (1:9, v/v)	>97	3037.1, 2.73	Condensed structure, no β−O−4′ aryl ether linkages, substantial β−β′ linkages, the presence of Hibbert's ketone	/	[93]
ChCl:LA (1:2) Poplar	130 °C, 1.5 h, 10%	Water; Ethanol/water mixture (1:9, v/v)	98	3929, 1.77	Low quantity of β-aryl-ethers with no recondensation, and an abundance of phenolic OH groups	/	[101]
ChCl:LA (1:2) Walnut endocarp cells	145 °C, 6 h, 10%	Ethanol/water mixture (1:9, v/v); Ethanol/water mixture (1:9, v/v)	92.1	4200, 1.8	Condensed G structure, little S and H units, absence of β–O–4′ and β-5′, increase of β–β′, and the formation of Hibbert's ketone	A broader but lower DGT peak, significantly higher residue fraction, lower thermal volatility and higher stability	[95]
LA Eucalyptus	110 °C, 6 h, 10%	Acetone/water (1:1 v/v); Acetone/water (1:1 v/v)	/	1660, 1.40	Less condensed	/	[54]
ChCl:LA (1:5) Eucalyptus	110 °C, 6 h, 10%			1980, 1.40	More condensed		
ChCl:LA (1:10) Eucalyptus	90 °C, 6 h, 10%			1990, 1.28	Trace amount of β–O–4, diminished β-β and β-5 linkages, less condensed		
	100 °C, 6 h, 10%			1825, 1.38			
	110 °C, 6 h, 10%			1790, 1.37			
	120 °C, 6 h, 10%			2080, 1.40			
	130 °C, 6 h, 10%			2050, 1.42			
Milled wood lignin Poplar			/	/	Low content of phenolic OH groups, no damage of etherified structures	/	[56]
ChCl:LA (1:9) Poplar	100 °C, 3 h, 2%	Deionized water; no purification	93		Low content of β–O–4 linkages, high content of phenolic OH groups, no phenylcoumarin group, presence of Hibbert's ketone	/	
	110 °C, 3 h, 2%		92				
	120 °C, 3 h, 2%		92.5				
	130 °C, 3 h, 2%		93				
	100 °C, 6 h, 2%		83				
	110 °C, 6 h, 2%		98				
	120 °C, 6 h, 2%		97				
	130 °C, 6 h, 2%		96				
ChCl:LA (1:10) Willow	120 °C, 6 h, 3.3%	Deionized water; Ethanol/water mixture (1:9, v/v)	90.02	1806, 1.34	Total phenol OH: 1.96 mmol/g	T_{MDR}: 384 °C; Solid residue: 38.1%	[88,91]
	120 °C, 9 h, 3.3%		92.37	1454, 1.20	/	/	
	120 °C, 12 h, 3.3%		94.46	1261, 1.18	High content of phenol OH group (2.05 mmol/g), nanoparticle lignin, less damaged structure, no β–O–4 linkage, dominant G	T_{MDR}: 380 °C; Solid residue 41.4%	
	120 °C, 18 h, 3.3%		95.08	1106, 1.15	Total phenol OH group: 1.68 mmol/g	T_{MDR}: 384 °C; Solid residue: 42.1%	
	120 °C, 24 h, 3.3%		95.40	1042, 1.13	Total phenol OH group: 1.59 mmol/g	/	
Double enzymatic lignin Eucalyptus			/	15770, 2.07	Phenolic OH: 0.65 mmol/g	A higher pretreatment temperature led to a reduction of β−O−4 content and more condensed structure units, G-type β−O−4 was lower than that of S-type	[90]
ChCl:LA (1:10) Eucalyptus	60 °C, 6 h, 5%	Hot deionized water (pH = 2.0);		12930, 1.89	Phenolic OH: 0.60 mmol/g		
	80 °C, 6 h, 5%	Hot deionized water (pH = 2.0)		8670, 1.75	Phenolic OH: 0.96 mmol/g		
	100 °C, 6 h, 5%			2240, 1.62	Phenolic OH: 1.45 mmol/g		
	120 °C, 6 h, 5%			2120, 1.45	Phenolic OH: 2.84 mmol/g		
					Phenolic OH: 3.13 mmol/g		

(*continued on next page*)

Table 2 (*continued*)

DES and biomass type	Reaction conditions	Precipitation and purification solvents	Purity	Structure			Reference
				M_W and DPI	NMR	Thermal	
	140 °C, 6 h, 5%			2070, 1.42			
ChCl:LA (1:1) Oil palm empty fruit bunch	120 °C, 8 h, 10%	Water/ethanol mixture (2:1); water	78.89	/	/	T_{MDR}: 254 °C,	[86]
ChCl:LA (1:15) Oil palm empty fruit bunch			80.64		Highest amount of β–O–4′ linkage and methoxyl group	T_{MDR}: 342 °C, Char residue 37%; Released higher concentration of gaseous products, more *p*-hydroxyphenyl- and guaiacyl-based aromatic compounds	
ChCl:FA (1:2) Oil palm empty fruit bunch			85.84		More condensed structure	T_{MDR}: 274 °C, Char residue 42%, less demethoxylation	
ChCl:LA (1:2) Corn straw	120 °C, 6 h, 5%	Deionized water; Water/ethanol mixture (9:1 v/v)	89.15	1977, 2.02	Phenolic OH: 1.469 mmol/g; β–O–4′: 8.37%	T_{MDR}: 354 °C, Char residue: 39.80%	[89]
betaine:LA (1:2) Corn straw			88.19	1985, 2.31	phenolic OH: 0.675 mmol/g; β–O–4′: 9.08%	T_{MDR}: 346 °C, Char residue: 36.04%	
Enzymatic mild acidolysis lignin Castor seed coats			/	5900, 2.2	Benzodioxane, resinol and cinnamyl alcohol end-units	/	[58]
ChCl:LA (1:2)	160 °C, 8 h, 10%; Castor seed coats	HCl aqueous solution (pH = 2); Ethanol/water (v/v, 1:9)	91	3420, 1.8	Major benzodioxane unit, and resinol unit, no cinnamyl alcohol end-unit		
ChCl:FA (1:2)			85	3120, 1.7			
ChCl:AA (1:2)			88	2810, 1.7			
ChCl:OA (1:2)			88	2440, 1.4			
ChCl:MA (1:1)			86	1810, 1.3			
ChCl:EG (1:2)			86	2330, 1.4			
ChCl:glycerol (1:2)			87	2510,1 1.5			
ChCl:LA (1:2) Castor seed coats	140 °C, 8 h, 10%		/	2410, 1.5			
	160 °C, 8 h, 10%			3420, 1.8			
	180 °C, 8 h, 10%			3430, 2.0			
	200 °C, 8 h, 10%			3150, 2.0	Only benzodioxane unit		
Enzymatic mild acidolysis lignin Poplar			/	5486, 1.89	Phenolic OH: 2.60 mmol/g; intense strength of β–O–4	T_{MDR}: 341 °C; Solid residue: 33%	[65]
ChCl:LA (1:2) Poplar	140 °C, 3 h, 5%	Deionized water; Deionized water		4058, 1.51	Phenolic OH: 4.60 mmol/g; decreased signal strength of β–O–4	T_{MDR}: 373 °C, Solid residue 53%	
ChCl:GA (4:1) Poplar				4369, 1.47	Phenolic OH: 4.00 mmol/g; decreased signal strength of β–O–4	T_{MDR}: 358 °C, Solid residue 47%	
ChCl:GA (3:1) Poplar				4225, 1.43	Phenolic OH: 4.83 mmol/g; decreased signal strength of β–O–4	T_{MDR}: 360 °C, Solid residue 50%	
ChCl:GA (2:1) Poplar				3382, 1.66	Phenolic OH: 5.89 mmol/g; decreased signal strength of β–O–4	T_{MDR}: 370 °C, Solid residue 51%	
Milled wood lignin Poplar			/	6838, 1.50	Typical presence of S-type and G-type β–O–4 bonds, low content of S-type OH	/	[29]
ChCl:glycerol: Al_3Cl (1:2:0.28) Poplar	120 °C, 4 h, 5%;	Acetone/water (70:30 v/v); Deionized water	94	5466, 1.67	No β–O–4 linkages; presence of C–C signals (β–β and β–5); increased S-type OH and decreased aliphatic OH; high content of phenolic OH groups	/	
BTMAC:LA (1:2) Corncob	100 °C, 2 h, 5%	Neutral Water; No purification	96.82	9770, 1.93	Presence of β–O–4 linkages, high content of ferulate and *p*-coumarate, presence of tricin	Solid residual: ~50%	[72]
	120 °C, 2 h, 5%		97.54	~5800;/		Solid residual: ~32%	
	140 °C, 2 h, 5%		97.82	3840, 1.97		Solid residual: 21.0%	
BTEAC:LA (1:2) Corncob	100 °C, 2 h, 5%		96.83	~11000;/		Solid residual: 60.6%	
	120 °C, 2 h, 5%		97.39	~7500;/		Solid residual: ~35%	
	140 °C, 2 h, 5%		97.89	~5000;/		Solid residual: ~25%	
TEBAC:LA (1:9) Wheat straw	100 °C, 10 h, 1:15	Deionized water; Deionized water	75.69	/	Mainly composed of the S units and the G units, no β–O–4′ linkage, formation of the Hibbert ketone structure	/	[86]
Cellulolytic enzyme lignin Switchgrass				~9500;/	β–O–4: 46.7%; S/G 0.70	Highest volatility and lowest solid residue	[27]
ChCl:EG + H_2SO_4 (1:2 + 0.5%)	110 °C, 0.5 h, 10%		78.66	~4500;/	β–O–4: 9.8%; S/G 0.77	Lignin extracted at higher temperatures or using a higher	

(*continued on next page*)

Table 2 (*continued*)

DES and biomass type	Reaction conditions	Precipitation and purification solvents	Purity	Structure			Reference
				M_W and DPI	NMR	Thermal	
ChCl:EG + H_2SO_4 (1:2 + 1.0%)		Acetone/water mixture (50:50, v/v)	82.94	~3900;/	β–O–4: 3.8%; S/G 1.01	acid concentration had higher thermal stability and lower volatility; low yields of phenolic monomers	
ChCl:EG + H_2SO_4 (1:2 + 0.5%)	130 °C, 0.5 h, 10%		84.62	~4200;/	β–O–4:/; S/G 1.17		
ChCl:EG + H_2SO_4 (1:2 + 1.0%)			88.35	~4000;/	β–O–4:/; S/G 0.85		
ChCl:EG + H_2SO_4 (1:2 + 0.5%)	150 °C, 0.5 h, 10%		86.90	~4100;/	β–O–4:/; S/G 1.33		
ChCl:EG + H_2SO_4 (1:2 + 1.0%)			87.22	~4150;/	β–O–4:/; S/G 1.02		
ChCl:EG + H_2SO_4+water (1:2 + 1.0% + 20%)	110 °C, 0.5 h, 10%		86.46	~4600;/	β–O–4: 24.4%; S/G 0.94		
ChCl:EG + H_2SO_4+water (1:2 + 1.0% + 20%)	130 °C, 0.5 h, 10%		91.36	~4750;/	β–O–4:/; S/G 1.07		
ChCl:EG + H_2SO_4+water (3:2 + 0.5% + 20%)	110 °C, 0.5 h, 10%		81.52	~3600;/	β–O–4: 40.3%; S/G 1.07		
ChCl:EG + H_2SO_4+water (3:2 + 1.0% + 20%)			87.02	~3000;/	β–O–4: 31.0%; S/G 1.07		
ChCl:EG + H_2SO_4+water (3:2 + 0.5% + 10%)			86.93	~3200;/	β–O–4: 33.7%; S/G 1.07		
ChCl:EG + H_2SO_4+water (3:2 + 1.0% + 10%)			90.42	~3000;/	β–O–4: 15.0%; S/G 1.07		
ChCl:tartaric acid: LA (1:1:4) *Pinus pinaster* Ait	150 °C, 2 h, 10%	Water; No purification	89	/	Existences of β–O–4 linkages and phenolic OH groups	Main weight loss at 217 °C and 360 °C	[37]
ChCl:OA dihydrate (1:1) Poplar	80 °C, microwave heating 3 min, 5%	Acetone/water (1:1 v/v); Acetone/water (1:1 v/v)	93.7–96.3	913, 1.25	Reduced H unit, no cinnamaldehyde end groups, no phenylcoumaran substructures, small S/G ratio (0.69), condensed structure	Suitable for aromatic chemicals production	[60]
ChCl:glycerol: H_2SO_4+20% water Switchgrass	120 °C, 1 h, 10%	Acetone:water (1:1 v/v); water	/	/	Preservation of β–O–4and similar with native lignin, suitable for monomer production, no cinnamate unit	/	[26]
Cellulolytic enzyme lignin Poplar			/	16733, 4.2	β–O–4: 69.2%; S/G ratio: 2.1; phenolic OH: 0.82 mmol/g	/	[63]
ChCl:PB (3:2) Poplar	160 °C, 3 h, 10%	H_2SO_4 aqueous solution (pH = 2); No further purification	/	1414, 1.6	β–O–4: 8.9%; S/G ratio: 2.1; condensed lignin: 39.3%; phenolic OH: 2.67 mmol/g		
	160 °C, 9 h, 10%			1180, 1.4	β–O–4: 6.4%; S/G ratio: 2.0; condensed lignin 52.5%; phenolic OH: 2.76 mmol/g		

the purity is also affected by the constituents in DESs to a certain extent. The purity of TEBAC:LA-extracted lignin initially increased with the increasing of molar ratio, reached the maximum (77.08%) at 1:11, and then decreasing at 1:13 [86]. The purities of lignin extracted by LA-based DESs were lower than that of FA-based DESs. This was attributed to the fact that there was lower efficiency for LA in breaking LCC linkages than FA [87].

The purity of lignin is usually affected by the reaction conditions. The search for an optimal lignin fractionation scheme for high purity is recommended by many researchers. The DES pretreatments generally involve reaction temperatures of 80–160 °C and duration time of 0.5–12 h (Table 2). The purity of DES lignin increased with the severity of treatment conditions [88]. Another attempt to increase its purity involved in the use of DES under aqueous conditions [25,27]. This is plausible because the presence of water reduces the interaction of EG with lignin and decreases the amount of residual EG in the recovered lignin [25]. Another possible explanation could be that the addition of water reduces DES viscosity, which in turn dissolves more carbohydrates from lignin into DES [27]. Other than DESs and pretreatment conditions, the adoption of anti-solvents during the recovery process of lignin also impacts on its purity. As summarized in Table 2, the commonly used anti-solvents are water (neutral or acidified by HCl or H_2SO_4 to pH 2), ethanol, or mixtures of ethanol/water and acetone/water. After the recovery of lignin, a purification process is normally adopted to further increase the purity of lignin. Alvarez-Vasco and co-workers first adopted a mixture of water/ethanol (volume ratio of 9:1) as an anti-solvent for lignin precipitation [32]. By using the same washing solvent, diverse DES lignin with high purity (>90%) has been obtained from various

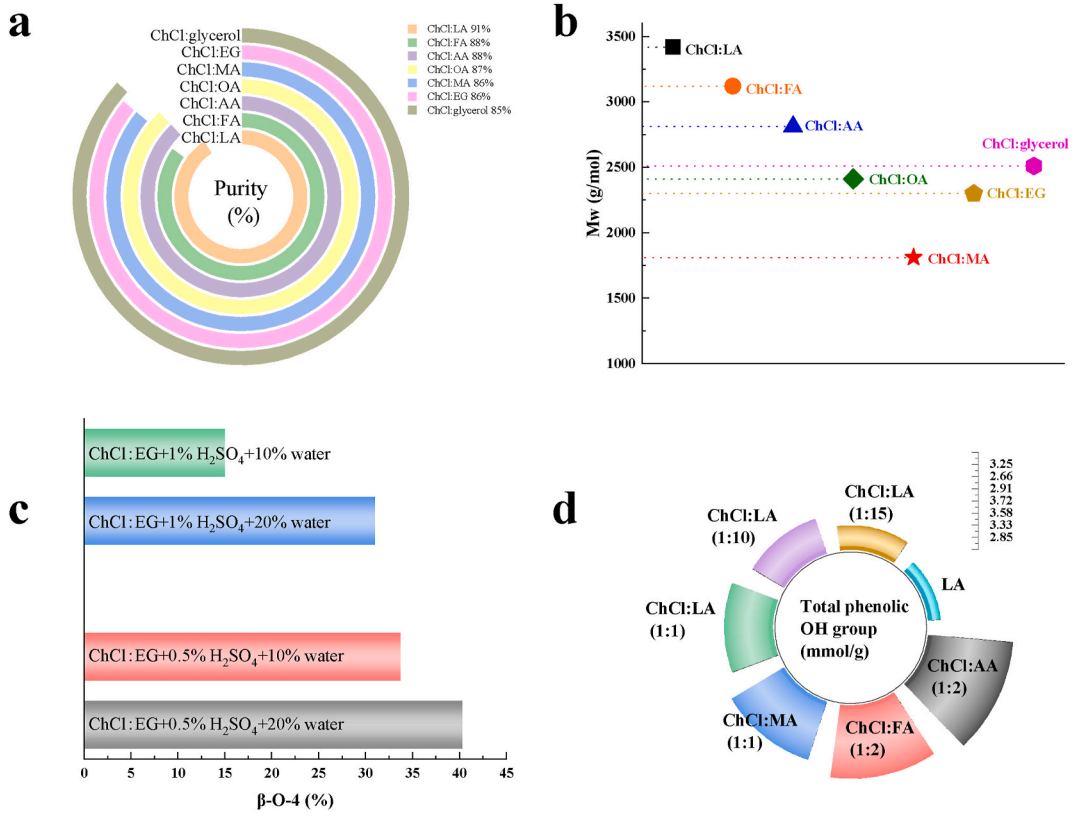


Fig. 6. Effect of DESs on the properties of DES lignin. (a) purity [58], (b) molecular weight [58], (c) content of β-O-4 bonds [27] and (d) content of total phenolic OH groups [50].

biomass. Acetone/water (50:50, v/v) is an alternative choice because this mixture is miscible with many DESs, and acetone can be easily removed via vacuum evaporation or recovered via distillation because of its low boiling point.

3.2. Molecular structure

The molecular structure of DES lignin, including molecular weight, the abundance of β–O–4 linkages and phenolic OH groups, as well as thermal stability and volatility, directly determine its susceptibility to further upgrading.

3.2.1. Molecular weight and distribution

Molecular weight and distribution are important properties of lignin. They correlate with the thermal stability, pyrolysis properties, polymer properties, and antioxidant activity. The molecular weight of lignin is positively related to the amount of ether linkages and inversely related to the degree of condensation [54]. Three indices are used to characterize the molecular weight and distribution of DES lignin: weight-average molecular weight (Mw), number-average molecular weight (Mn) and polydispersity (Mw/Mn, PDI). As shown in Table 2, DES lignin normally has a lower (smaller Mw and Mn) and narrower (smaller PDI) distributed molecular weight compared to native lignin, such as cellulolytic enzyme lignin (CEL), enzymatic/mild acidolysis lignin (EMAL), and milled wood lignin (MWL). This feature is favorable for its further valorization. The molecular weight of lignin is a result of depolymerization and condensation acting in concert, whereby the molecular weight of lignin decreases along with the depolymerization by DESs, and the depolymerized lignin condenses when the pretreatment conditions are too severe.

HBDs and HBAs greatly influence the molecular weight of lignin. As shown in Fig. 6b, the Mw of lignin is remarkably affected by the DESs used [58]. The Mw and PDI of lignin isolated by ChCl:LA were lower than those of betaine:LA. This indicates that ChCl:LA has a stronger effect on lignin dissociation and tended to form more homogeneous lignin [89]. Apart from DESs, the molecular weight of lignin greatly decreases with the increasing severity of pretreatment conditions, such as increased temperature [90] and reaction time [91]. However, a contrary observation was reported, which described how the Mw of ChCl:LA-extracted lignin from castor seed coats increased from 2410 g/mol to 3420 g/mol as the temperature increased from 140 °C to 160 °C [58]. This partly accounted for additional cleavage of ester linkages between hydroxycinnamic acid and lignin under higher temperature. Therefore, the molecular weight and distribution of lignin is a combined action of depolymerization and repolymerization, which jointly depends on the botanical origin of the biomass, the DES constituents, and reaction severity.

3.2.2. Abundance of β−O−4 linkages

The abundance of β−O−4 linkages is a key factor that governs the further valorization of lignin. The preservation of the β−O−4 linkages is critical for the high-yield production of aromatic monomers via lignin depolymerization. It has been reported that the theoretical monomer yield is roughly proportional to the square of the fraction of cleavable inter-unit ether bonds (i.e., β−O−4) [92]. Several catalytic conversion approaches targeting the β–O–4 cleavage for aromatic monomer productions have been developed (see Section 6). Lignin with low amounts of β−O−4 linkages generally exhibit satisfactory thermal stability due to the absence of liable ether bonds, and is suitable for polymer production. Characterization and understanding of the structure-process-property relationship among DESs, process conditions and DES lignin are necessary. Generally, the amount of β−O−4 linkages of DES lignin is influenced by the constituents of DES, reaction conditions, and the addition of water.

DES constituents play pivotal roles in the abundance of β–O–4 linkages. Carboxylic acid-based DESs (such as ChCl:LA) normally lead to the

complete cleavage of β–O–4 bonds [32,88,93], whereas acidified, aqueous polyalcohol-based DESs (such as ChCl:glycerol) are able to partially preserve β–O–4 bonds to a great extent [25–27]. It was observed that ChCl:LA broke more β–O–4 linkages than betaine:LA [89]. The β–O–4 linkages in lignin are more susceptible to breakage as the amount of ChCl increases in the DES system [54]. The cleavage of intermediates formed between chlorine ions and lignin model compounds, aided by the presence of a strong acid counterpart, accelerates the cleavage of β–O–4 [68,86].

Tailoring lignin with varying amounts of β–O–4 linkages can be tuned by adjusting reaction conditions. As shown in Table 2, the amounts of β−O−4 ether bond linkages remarkably decrease with increasing temperature [27,54,94]. A higher pretreatment temperature favors a reduction in the amount of β–O–4 bonds. When a higher temperature (e.g., 145 °C) is adopted, complete cleavage of β–O–4 bonds occurs during lignin fractionation, regardless of botanical origins [32,93,95]. The breakage rate of β–O–4 is the fastest among all the lignin linkages due to the bond energy being the lowest. Compared with temperature, treatment time has minor effects on β–O–4 linkages. For example, when extracting lignin at 90 °C, considerable amounts of β–O–4 bonds (115.3 mg/g) were preserved even after pretreatment for 168 h, which was slightly lower than that of 24 h (137.5 mg/g) [94]. Additionally, solid loading also affects the amounts of β–O–4 linkages. High solid loading is recommended because of the relatively higher amounts of β–O–4 bonds [25].

Lignin with varying abundances of β–O–4 linkages can be obtained by the addition of water. The Wan research group has reported that H_2SO_4-acidified ChCl:glycerol and ChCl:EG are able to preserve β–O–4 bonds to a great extent [25–27]. H_2SO_4 plays a pivotal role in lignin fractionation because it initiates the cleavage of β–O–4 linkages by protonation of the Cα-position. Upon the introduction of water, the catalytic activity of acid protons is decreased due to their solvation by water molecules [27]. Varying the amount of water results in the content of β–O–4 linkages varying accordingly. As seen in Fig. 6c, the greater the amount of water added, the greater the number of β–O–4 linkages that were preserved [27]. Generally speaking, lower temperature, shorted time, higher solid loading, or addition of water give rise to a higher percentage of preserved β–O–4 bonds in DES lignin. However, such passive stabilization of β–O–4 linkages generally occurs at the expense of lignin fractionation yield and purity. Active stabilization by using capping agents (such as formaldehyde, acetaldehyde, or propionaldehyde) to protect α, γ-diol in lignin has recently been proposed because it does not compromise lignin extraction efficiency [96,97]. It is worth exploring chemicals that can serve as both DES constituents and capping agents, as well as developing processes for the fractionation of uncondensed or well-preserved ether-bonded DES lignin.

3.2.3. Abundance of phenolic hydroxyl groups

The abundance of phenolic OH groups, which is a reflection of its reactivity, plays an important role in downstream valorization. During the fractionation of lignin by DESs, the amount of phenolic OH groups can be increased by the breaking of β–O–4 bonds [32,56] and the demethylation of methoxy groups within lignin units [91]. Appreciable amounts of phenolic OH have been identified in DES lignin through FTIR and ^{13}C or ^{31}P NMR spectroscopy. Generally, there is a higher number of phenolic OH groups in DESs-extracted lignin as compared to native lignin (Table 2), which is mainly due to the cleavage of β–O–4 bonds.

As shown in Fig. 6d, there is a large influence on the phenolic OH groups by the constituents and their molar ratios within DESs. Treatment temperature and time also affects the number of total phenolic OH groups in DES lignin [63,88,98]. In addition, plentiful phenolic OH groups contributed to their H-binding with residual DES due to the increased affinity between them [32]. A small amount of EG was found remaining in the ChCl:EG-extracted lignin because EG was able to react with lignin by forming esters and ketals [25,27]. The phenolic OH groups probably decreased by subsequent condensation or repolymerization when treatment conditions were inappropriate. The abundance of condensed lignin increased as temperature increased [30] or as a result of extending the pretreatment time [63]. Therefore, the addition of stabilization/capping agents will prevent condensation of phenolic OH groups [96,97].

3.2.4. Elemental composition

Normally, the elemental composition of lignin (mainly carbon, hydrogen, and oxygen) is measured using an elemental analyzer or XPS. The amounts of each element are related to the chemical structure and purity of lignin. The percentage of carbon and hydrogen elements are generally higher in DES lignin than their original feedstock [42], whereas they are similar to native lignin [89]. The amount of oxygen in DES lignin is impacted by DES constituents [86] and fractionation conditions [88]. The oxygen-containing functional groups or bonds (such as methoxy group and aryl ether linkages) in lignin are easily eliminated with increasing reaction severity. In addition, elemental analysis, combined with molecular weight, can be used to calculate the polymerization degree and empirical formula of DES lignin [89].

3.3. Thermal characteristics

Thermal characteristics of lignin mainly refer to thermal stability and thermal volatility. It is important to understand the relationship between the chemical structure and properties of lignin to further valorize lignin. The thermal properties of lignin can be characterized by TGA, TG-FTIR, and Py-GC/MS. Usually, decomposition temperature and curve, temperature of maximum degradation rate (T_{MDR}), solid or char residue, and pyrolysis product distribution are used to describe its thermal properties. Compared to native lignin, DES lignin decomposes at higher temperatures and yields less total aromatic monomers. These features are due to the partial breakage of liable ether bonds during DES fractionation. Because the aromatic side chains are partially cleaved during DES fractionation, simple and short-side chain phenols are more selectively produced from DES lignin.

The DES lignin isolated by different DESs is characterized by different T_{MDR} and solid residues. The T_{MDR} and solid residue of lignin extracted by ChCl:LA are higher than those of betaine:LA [89], ChCl:GA [65], and ChCl:FA [86]. These indicate that ChCl:LA-isolated lignin presents a more condensed structure with greater stability. In addition, the molar ratio of HBA and HBD plays a significant role in tailoring lignin thermal stabilities. The solid residue of DES lignin isolated by ChCl-based DESs increases with the increasing HBD [27,65]. Furthermore, the addition of water affects the thermal stability of DES lignin. There was increased thermal stability for lignin fractionated by aqueous ChCl:EG in comparison to its counterpart lignin [27]. Moreover, thermal stabilities are also affected by the solid loading [25], fractionation temperature [70] and time [91]. Additionally, the thermal stability of DES lignin corresponds well with its molecular weight.

During pyrolysis, lignin is decomposed into phenolic monomers and oligomers. Both of them can be used for polymer synthesis or catalytically deoxygenated to produce biofuels. TG-FTIR showed that the different gas evolution trends for three DES lignin were rather similar. This observation suggests that DES lignin exhibits the potential to produce syngas and phenolic compounds by gasification, pyrolysis [86], or torrefaction [99]. As expected, the pyrolysis products of DES lignin are mainly aromatic compounds [86]. The thermal volatility of DES lignin is affected by DESs and fractionation conditions. Lignin extracted by ChCl:LA released a higher concentration of gaseous products, which was possibly associated with its higher percentage of the β–O–4 linkages. Analysis of the abundance of *p*-hydroxylphenyl- and guaiacyl-based aromatic compounds revealed that more optimal de-methoxylation performance was observed for ChCl:LA-extracted lignin as compared to ChCl:FA-extracted lignin [86]. Py-GC/MS showed that the pyrolysis products of DES lignin at different extraction times were similar, but with quite different relative amounts [91]. Fractionated lignin at high

solid loading improved the quality of DES lignin in terms of volatilization and upgrading for phenolic monomers [25].

Different lignin samples with varying thermal stability and volatility are used in different applications and have different values. Generally speaking, lignin with higher thermal stability and higher solid residue is preferred in composite materials and other bio-based carbon applications. In contrast, lignin with higher volatility and lower solid residue is desired as a precursor feedstock of biofuels and chemicals. Further investigations are recommended to identify the factors controlling the selectivity of pyrolysis products and to tune lignin thermal properties for tailored upgrades.

3.4. Other properties

Apart from the above-mentioned properties, DES lignin also possesses many biological properties. As with other technical lignin, DES lignin exhibits some merits, such as blocking ultraviolet light and scavenging free radicals, as well as anti-microbial activity. These biological properties are derived from the presence of the phenylpropane structure and phenolic hydroxyl groups [100]. DES lignin exhibits stronger antioxidant activity than native lignin. The antioxidant activity of DES lignin is closely related to its amounts of tricin substructure [70] and phenolic OH groups [71]. Furthermore, the antioxidant functionality of DES lignin is affected by the molar ratio of DESs [101] and the severity of treatment conditions [55].

DESs are conducive for the formation of nanoparticles due to the fast nucleation reaction catalyzed by the DES system. DES lignin exhibits homogeneous morphology, nanosized scale, and uniform distribution. Lignin nanoparticles with different sizes, distributions, and structures can be achieved by adjusting DES constituents and reaction conditions [102]. Generally, high severity of treatment conditions decreases the size of DES lignin nanoparticles due to the depolymerization of lignin. However, some researchers reported that with increasing temperature, the particle size of lignin showed no significant variation [90] or even increased [94]. This discrepancy may originate from the difference in the structure of lignin originating from various biomasses. Different lignin sources exhibit varying sensitivities to the severity of DES fractionation. The incorporation of DES lignin nanoparticles confers materials with UV-blocking, antioxidant properties and improved mechanical performance [103].

Collectively, DES lignin exhibits high purity, tunable molecular structure and thermal characteristics, and appealing biological properties. It is a good candidate for next-generation functional nanocomposites, such as active packaging materials applied in the medicine and food industry. Therefore, as a new class of technical lignin, the properties of DES lignin should be comprehensively characterized, and its application potential should be fully tapped in the future.

4. Lignin solubilization in DESs

Lignin dissolution is a prerequisite for lignin fractionation and valorization in a homogeneous phase. Generally, the dissolution of lignin in a solvent involves disrupting π–π stacking interactions between lignin aromatic groups and establishing new H-bonds between solvent and lignin [104]. Four types of technical lignin are usually used to evaluate the solubility of lignin: enzymatic hydrolysis lignin (EHL), organosolv lignin (OL), dealkaline lignin (DAL), and sodium lignosulfonate (SL). In spite of conventional organic solvents being able to dissolve some technical lignin, their low vapor pressure limits their applications. However, the low volatility and greenness of DESs could overcome the drawbacks of volatile solvents [4]. For example, the solubilities of DAL in ChCl:LA (1:10) at 60 °C are 17 wt% [105]. More recently, a fast and simple methodology using FTIR spectroscopy to assess the solubility of lignin or lignin-derived compounds has been recently developed [106]. This method will provide many conveniences for the selection of DESs that highly dissolve lignin.

4.1. Mechanisms of lignin solubilization in DESs

Currently, two mechanisms explaining the good solubility of lignin in DESs are presented, namely, the hydrotropic mechanism and the co-solvency mechanism. The hydrotropic hypothesis was confirmed by two observations: (1) the shape of solubility curves with strong maxima at intermediate compositions between pure water and neat DES; (2) the presence of nanoaggregates typical of the density fluctuation of the hydrotropic solubilization, as ascertained by the use of dynamic light scattering [107]. Similar observations that the addition of water to proline:malic acid [73] and betaine:lysine [108] systems increased lignin solubility also support the hydrotropic hypothesis. The co-solvency hypothesis was corroborated by the findings that the most optimal solvent performances was obtained with neat ChCl:LA [109]. The hydroxyl group of LA in ChCl:LA increased the polarity of the DES system, which facilitated the dissolution of lignin, and the ChCl group formed a new connection with the phenolic fraction of lignin [105]. The solubilities of lignin in various DESs can be partly explained by the difference in interaction energy ΔE (calculated based on π–π stacking and H-bonding interactions) between lignin and DES [110]. However, because the application of DESs for lignin dissolution is in its early stage, the underlying mechanism of lignin solubilization in DES systems is still poorly understood. Additional in-depth investigations should be conducted to provide further mechanistic insights in future works.

4.2. Effects of DESs on lignin solubilization

The capacity of DESs to dissolve lignin is significantly affected by their functional groups of HBAs. The solubilities of lignin in various DESs that are composed of different HBAs and ChCl significantly vary [57]. Same observation was made for in the LA-based DESs [111]. When containing the same halogen ions, betaine-based DESs only dissolved 12% lignin, while ChCl-based DESs dissolved 100% lignin [69]. Fig. 7 shows the lignin solubilities of 25 kinds of DESs, including type III (LA-based) and type V (DBU-based) DESs. Obviously, lignin solubility significantly varies based on the nature of DESs and the type of technical lignin. As shown in Fig. 7a, comparison of four DESs with different functional groups reveals that the lignin solubility follows the order: ATMAC > ChCl > BTMAC/BTEAC. Particularly, the most optimal dissolution of the four types of technical lignin was observed for ATMAC:LA. The highest solubilities of EHL, DAL, and OL in ATMAC:LA (1:2) at 70 °C were approximately 50 wt% [31]. Recently, it has been reported that technical lignin can be efficiently dissolved in type V DESs [20]. For example, the solubility of DAL in type V DBU:PEG (1:3) is as high as 48.4% (Fig. 7c). s.

The solubilization of lignin is also influenced by the chemical structure of HBDs. HBDs are found mainly responsible for lignin solubility in ChCl:LA, as determined through quantum chemistry calculations and molecular dynamics simulations. It is LA rather than chlorine ions that form the strong non-covalent interactions with the OH groups of a lignin model compound [110]. Lignin solubility decrease when two phenyl groups are present [40]. Higher lignin dissolution performance was obtained for ChCl:EG when tested against other DESs due to the presence of two OH groups [104]. The lignin solubilities at 60 °C in various DESs composed of ChCl and different HBDs are in the order: ChCl:LA > ChCl:OA dihydrate > ChCl:OA anhydrous [111]. As shown in Fig. 7, the lignin solubilities of various DBU-based type V DESs vary greatly depending on the nature of HBDs. For example, the EHL solubility of DBU:PEG (1:3) is remarkably higher than that of DBU:ε-caprolactam (1:3) (Fig. 7d).

Apart from the functional groups of HBAs and HBDs, other factors such as molar ratio and additives have influences on lignin solubility. When the molar ratio of ChCl:LA increased from 1:1.3 to 1:10, the lignin solubility steadily increased from 4.55 wt% to 11.82 wt% [111]. The EHL solubility of DBU:PEG (1:3) was remarkably higher than that of DBU:PEG (2:1) (Fig. 7d). Additionally, aqueous solution of DESs has

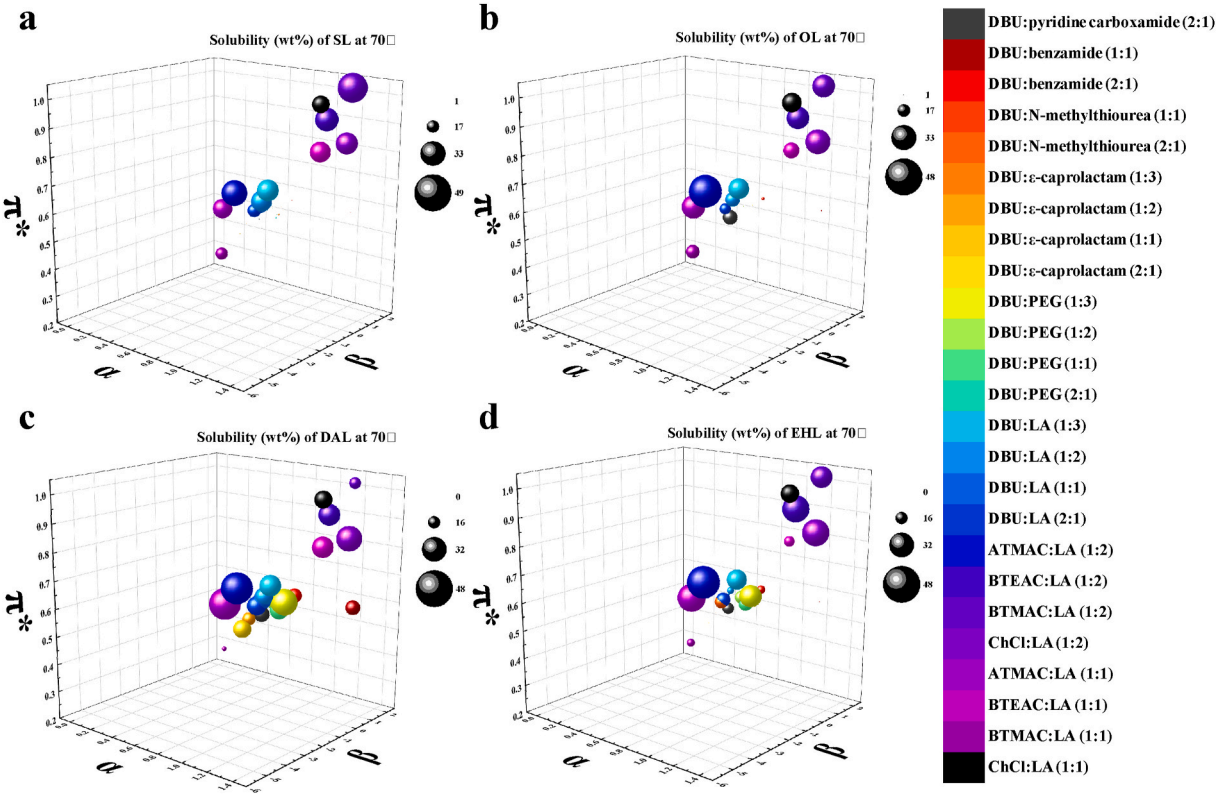


Fig. 7. The correlation between lignin solubility and the K–T solvatochromic parameters of DESs (the ball size represents the lignin solubility) [20,31].

been demonstrated as a new class of powerful solvents for lignin [107].

4.3. Relationship between Kamlet–Taft parameters and lignin solubility

Lignin solubility can be explained and predicted by the solvent properties of DESs, such as those defined by the K–T model. A high HBA capacity is typically required in order for the solvent system to effectively solubilize lignin [7]. To better understand the solubility of lignin in DESs, Liu and co-workers comprehensively analyzed the correlation between lignin solubility and DESs' K–T parameters (Fig. 7). An inductive analysis revealed that the α and β values were more closely related to lignin solubility than the π* value. In terms of type III DESs, their abilities to dissolve lignin positively correlate with the subtraction value of α and β [31]. As for type V DESs, the correlation between lignin solubility and K–T solvatochromic parameters is complicated [20]. Considering that the K–T parameters for DESs can be easily tuned and measured, the design of suitable DESs for the efficient dissolution of lignin from lignocellulosic biomass holds enormous potential.

4.4. Correlation between lignin dissolution and lignin fractionation

The dissolution of lignin by a DES does not certainly guarantee effective lignin fractionation from biomass. For instance, excellent solubilization of lignin was obtained with pionic acid:urea, but it was unable to cleave the β–O–4 ether linkage [68]. Furthermore, the addition of water has been reported to increase lignin dissolution in DESs [107]. However, a contrary phenomenon was also observed whereby 20% water addition to ChCl:glycerol reduced delignification efficiency [26]. These divergences suggest that the underlying mechanism of lignin dissolution in DESs might differ from that of lignin fractionation from lignocellulosic biomass by DESs. The fractionation of lignin from lignocellulosic biomass may involve the cleavage of various bonds in LCC, while lignin solubilization in a certain DES can simply follow possible hydrotropic or co-solvency mechanisms [4]. Because the fractionation of lignin from biomass and the solubility of lignin DES are still in their infancies, exact mechanisms of lignin fractionation and solubility remain elusive. Therefore, more efforts should be devoted to designing effective lignin-dissolving DESs, especially those that can perform well for lignin fractionation.

5. Computational approaches in lignin fractionation, dissolution, and conversion using DESs

The general theoretical and computational methods, namely conductor-like screening model for real solvents (COSMO-RS) model, quantum chemistry (QC) calculation, and molecular dynamics (MD) simulations, have been applied for studying the mechanisms of fractionation, dissolution and conversion of lignin. These methods describe intrinsic characteristics such as accurate structures, energies, and interactions, providing insightful views at the atomic level.

5.1. Conductor-like screening model for real solvents model

The COSMO-RS model is considered to be the most accurate model for predicting the properties of DESs. COSMO-RS can be used to calculate the thermodynamic equilibrium properties (such as activity coefficients, solubility, partition coefficients, vapor pressure, and free energy of solvation) and dominant interactions (H-bonds, misfits, and van der Waals force) in the DES system [112]. COSMO-RS is useful for the designation of new DES solvents and the verification of the potential for lignin dissolution. It is time-efficient and robust for prior screening of suitable DESs for target fractionation. COSMO-RS greatly alleviates the experimental burden. The extraction capacity of DESs can be quantified using the infinite dilution activity coefficient (γ^{∞}), σ profile, and σ potential. Through the prediction of partition coefficients, a range of solvent systems that may be suitable for the effective separation of lignin-derived monomers has been rapidly screened from a vast pool of solvents [113]. Apart from predicting the solid–liquid equilibria of

phenolic-based DESs [19], COSMO-RS can be used to study the molecular contributions that confer affinity between DES and lignin. COSMO-RS has been reported as a useful tool for predicting the solubility or fractionation of phenolic compounds such as rutin [114], quercetin [115] and hydroxytyrosol [116] in DESs. Thus, it is possible for COSMO-RS to provide guidance to screen the most effective DES for extracting and solvating lignin. Recently, COSMO-RS has been recognized as a promising tool for the designation and fast screening of emerging type V DESs [117]. Further work is anticipated to fully extend its application to the fractionation and valorization of lignin.

5.2. Quantum chemistry calculation

QC calculation is an effective and widely used method to study molecular structure. QC calculation can predict chemical reaction intermediates, reaction barriers, and other thermodynamic parameters via solving the basic Schrödinger equation [34]. Among various methods, the density functional theory (DFT) method is usually used to interpret the behavior of lignin in different DES environments. The M06–2X functional and solvation model density are of great scientific and technological importance for predicting the chemistry-driven effective interactions and supramolecular cooperative interactions in DES systems [110].

QC calculation can greatly assist in understanding the mechanism of lignin fractionation/dissolution mediated by DESs and supporting experimental results. Given that DES is a recent emerging solvent, the molecular structures of some DESs (e.g., ChCl:PB) are not available. Therefore, many researchers adopt QC calculation to gain insights into their structural information. By means of QC calculation, detailed information regarding H-bond and non-covalent interactions in ChCl: glycerol was identified [29], and the molecular structure of ChCl:PB was estimated [63]. QC calculation has also been used to reveal the intermolecular interactions of lignin, such as configurations, chemical reactivity, non-covalent interactions [110], and bond dissociation enthalpy [29,64,68]. For example, Ji and Lv revealed that the domain non-covalent interactions in lignin were H-bonds and π–π stacking [110]. Lastly, QC calculation offers valuable and complementary insights into the mechanism of lignin fractionation, dissolution, and conversion in DES environments. A comprehensive elucidation on the mechanism of lignin fractionation using a ternary DES has been made by Xia et al. [29]. They demonstrated that the H-bonding strength of OH···Cl is higher than that of OH···O in lignin, and multisite Cl^- from $AlCl_3$ is required for the efficient disruption of the H-bonding network in lignocellulose. Ji and Lv determined that strong O···H bonding and C–H···π contribute to the lignin dissolution in ChCl:LA [110]. Moreover, QC calculation is able to predict the modification and conversion pathways of technical lignin. The direct contribution of characteristic lignin subunits to antioxidant capacity was predicted based on the calculation of O–H bond dissolution enthalpy and electron transfer enthalpy [118]. QC calculation provides a new quantitative technique for increased understanding of the antioxidant activity of lignin. A proposed mechanism for lignin model compound cleavage was analyzed and supported by computing the free energy profiles of the first step in three acidic DESs. The authors also revealed that halide intermediates are important for speeding the cleavage reaction [68].

5.3. Molecular dynamics simulations

To overcome the limited scale of DFT, MD simulations are adopted to move from individual molecules to the bulk properties of more than thousands of atoms. MD simulations employ the classical Newtonian equation and provide insights for complex interactions at the mesoscale level. Currently, the most widely used force fields in DES systems include AMBER [33,110], CHARMM [43], OPLS-AA [74], GROMOS, and ReaxFF-RMD [119]. Recently, a new force field, called OPLS-DES, has been specially developed for DESs. OPLS-DES has shown high reliability in the reproduction of bulk-phase physical properties and local intermolecular interactions for 8 different ChCl-based DESs [120].

At present, MD simulations mainly focus on studying the relationship between the structure, and physical/chemical properties of DES, and the interaction between DES and biomass to observe the solubility of biomass models in DES. The Hildebrand and Hansen solubility parameters for DESs can be computed based on MD simulations. This strategy contributes to the selection of solvents for certain applications [33]. Additionally, MD simulations have been adopted to understand the composition effect on ChCl:EG at the microscopic level. Distribution functions show that the structures are governed by the balance of H-bond networks and the different populations of the EG molecule conformations. In the water-containing system, water competes for association with the anion [74]. Furthermore, MD simulations reveal that it is LA rather than Cl^- that provides strong non-covalent interactions with the hydroxyl groups of a lignin model compound VG [110]. More recently, a lignin−carbohydrate complex model (PG-LCC) in ChCl-based DESs has been studied by MD simulations [119]. By computing the cracking rate and production of phenolic compounds *in silico*, the effects of temperature, HBDs, chain length, acidity of acidic HBDs, and molar ratio on lignin fractionation have been elucidated. The mechanism of PG-LCC in DESs is basically understood and is roughly classified into four pathways. Moreover, MD simulations confirm that certain lignin bonds (e.g., β–O–4) are stretched under microwave irradiation, leading to an increased probability of breakage [43]. This finding partially explains the synergistic mechanism between DESs and microwaves in the fractionation of lignin at the molecular level. At present, exploration of the mechanism of lignin fractionation of lignocellulosic biomass in the DES system is in the initial stage.

5.4. Other methods

The integration of thermophysical properties with parameters that describe DESs in a human and machine-readable format is highly desirable. A standardized data exchange format to exchange, integrate, store, analyze, and publish data has been proposed and recommended [121]. Based on the Chemical Markup Language, the thermophysical properties of aqueous DESs were analyzed by machine learning. The trained model showed an excellent correlation with experimental and simulation data. The impacts of water content and temperature on the viscosity of DESs have been analyzed by meta-analysis workflow using Arrhenius and Vogel–Fulcher–Tammann–Hesse models [122]. More recently, multivariate analysis methods including principal component analysis and partial least squares analysis have been used to identify key factors from 42 parameters that influence lignin fractionation when using DESs [51].

In summary, computer-assisted modeling, computation, and simulations play key roles in the understanding of the efficient utilization of lignin. Although there is less investigation on intrinsic mechanisms from a theoretical viewpoint compared to experimental studies, the increase in the computational power and the availability of additional computational resources will undoubtedly promote mechanistic studies on lignin fractionation, dissolution, and valorization in DES systems.

6. Valorization of lignin in DES medium

Lignin is an appealing renewable biopolymer, and possesses numerous attractive properties such as high carbon content, thermal stability, biodegradability, absorbance of UV irradiation, and antioxidant and antimicrobial activities [123]. The valorization of lignin is essential to enhance the economic and environmental profitability of the overall lignocellulosic biomass conversion process. With accumulating interest in the fractionation of lignin using DESs, the produced DES lignin exhibits many desirable properties for further upgrading. To achieve the long-term goal of developing a cost-effective process for lignin depolymerization via DESs, it is important to explore strategies to

valorize lignin in DES systems or DES lignin. Currently, the valorization of lignin can be categorized into two groups: functionalization and depolymerization, as shown in Fig. 8.

6.1. Functionalization

As a natural polymer, lignin can be straightforwardly used as technical polymer to improve or change the properties of existing polymeric materials. For instance, lignin is used as an antioxidant and UV-shielding additive in plastics for active packaging [123]. However, the applications of lignin are limited due to its inferior mechanical properties, low reactivity, and incompatibility with other polymers. Functionalization of lignin through chemical modification renders lignin with different properties, and thus broadens its applications. Functionalization normally involves increasing the reactivity of targeted functional groups such as OH, methoxyl, carbonyl, carboxyl or aromatic groups via hydroxyalkylation, esterification, alkylation, amination, and epoxidation. Grafting from and grafting to are the most well-defined methodologies for lignin modification [123]. Such modifications include the introduction of new active sites on lignin by combining other materials, and the functionalization of the hydroxyl group to enhance its reactivity [100]. Through functionalization, lignin can be transformed into macromolecular or polymeric materials (Fig. 8), such as nanoparticles [102], carbon nanofibers [124], phenol-formaldehyde resins [125], grafted lignin [126], aerogels [127] and supercapacitors [128].

DESs are a new generation of reaction media for lignin functionalization. The usage of DESs can overcome the limited solubility of lignin in other media and contribute to its enzymatic modification [129]. The use of DESs for enzyme-catalyzed functionalization rather than other solvents (e.g., alkaline solution) is advantageous because they possess satisfactory biocompatibility with many enzymes. Apart from usage as reaction media, DESs can be used as agents to improve lignin reactivity via increasing phenolic OH groups. The $ZnCl_2$:LA-functionalized lignin showed enhanced reactivity toward the formation of phenolic formaldehyde adhesive. This strategy resulted in increased phenolic hydroxyl groups and decreased methoxy groups, and higher strength and thermostability for the lignin-phenol-formaldehyde adhesive compared to the control one [125]. After treating alkali lignin with ChCl:$ZnCl_2$, the reactivity of lignin with formaldehyde was enhanced, and the mechanical strength of the phenol-formaldehyde resin remained unchanged by replacing 40% phenol with DES-treated lignin [130]. After pretreatment with ChCl:FA, the structure of masson pine alkali lignin changed greatly. The modified lignin was suitable for substituting 100% substitution of phenol in the preparation of highly-ordered lignin-derived mesoporous carbon [124].

Lignin-based smart materials, including (1) sensors and biosensors (lignin-based carbon and graphene quantum dots for sensing applications), (2) nano/microparticles and hydrogels that encapsulate and release compounds under the application of specific stimuli, and (3) shape-programmable materials, can be fabricated for current and future applications [123]. More recently, lignin has been viewed as a promising material for the production of DESs-conductive gel. Lignin fiber gels can be simply prepared by using DES as the electrospinning solvent [127]. A new ionic gel can be rapidly developed by using a dynamic redox reaction between lignin and $Fe_2(SO_4)_3$ in betaine:EG Ref. [128]. Their potential application in supercapacitors and wearable sensors has been demonstrated. Nevertheless, hitherto, no study has been conducted on the use of DES lignin as a precursor for smart materials production.

Importantly, the selection of the type of lignin is strongly related to the intended application. Normally, DES lignin is highly pure, and therefore, it can be used for high-performance materials, such as those used in biomedical applications. Besides, DES lignin could be a promising substrate for polymer synthesis without further being functionalized due to its high reactivity *in situ* endowed by DES fractionation. For instance, DES lignin fractionated by ChCl:LA showed higher amounts of phenol OH groups and more optimal reactivity [89] compared to other technical lignin. Therefore, there is a huge amount of interest in using DES lignin in various applications via functionalization.

6.2. Depolymerization

The other valorization process for lignin is depolymerization. As mentioned above, the isolated lignin obtained from various DESs exhibits many desirable properties for further upgrading. The depolymerization of lignin can currently be achieved via the biological route, the thermochemical route, and the electrochemical route (as shown in

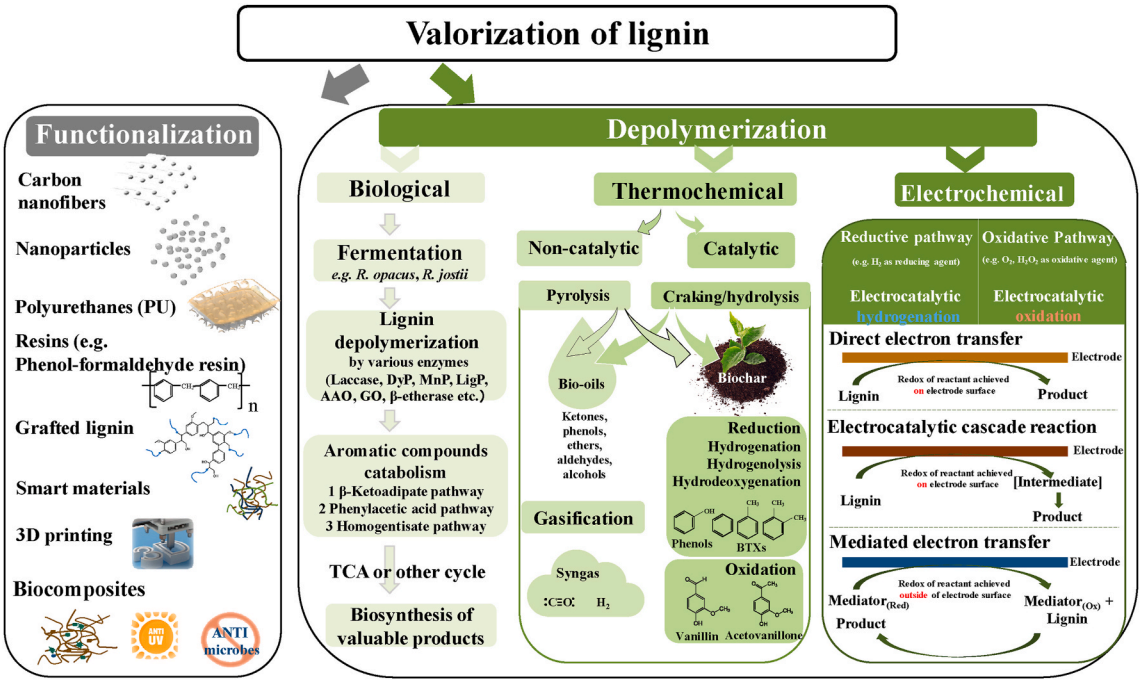


Fig. 8. Various strategies for lignin valorization.

Fig. 8).

6.2.1. Biological route

In nature, lignin can be degraded by many ligninolytic microorganisms. Many fungi (e.g., white rot fungi) [39] and bacteria (e.g., *Rhodococcus opacus*) [131] are able to efficiently delignify lignocellulosic biomass. Lignin can be *in vivo* depolymerized by a broad enzymatic machine into aromatic compounds. The enzymatic arsenal includes laccase, manganese peroxidase (MnP), versatile peroxidase (VP), heme-dependent lignin peroxidase (LigP), dye-decolorizing peroxidase (DyP), β-etherase, aryl alcohol oxidase, and glyoxal oxidase. Subsequently, these compounds are metabolized via the β-ketoadipate, phenylacetic acid, and homogentisate pathways, and finally enter into the TCA or another cycle to biosynthesize various valuable products [131].

It has been reported that *Pandoraea* sp. B-6 can assist in the further depolymerization of lignin remaining in the solid residues from ChCl:LA-pretreated rice straw [132]. FTIR and HSQC NMR results revealed that *Pandoraea* sp. B-6 further promoted the fragmentation of the G units in lignin after DES pretreatment. A more recent study demonstrated that the DES waste extraction liquor from lignin could be converted into microbial lipid by *Rhodotorula glutinis* fermentation [46]. Additionally, metabolic engineering and synthetic biology are usually used to tailor and improve the ligninolytic microbe's ability to degrade and metabolize lignin [131]. Despite noteworthy development, the biological upgrading of lignin is still challenging because the overall efficiency is relatively low. The biological conversion of lignin normally is very time-consuming (a few days or weeks) and inefficient as compared to other depolymerization routes.

Apart from the microbial depolymerization, enzymatic depolymerization of lignin is a promising alternative strategy. Strikingly, some laccases exhibit satisfactory biocompatibility with many DESs. The activity of *Trametes versicolor* laccase was preserved, and particularly enhanced up to 200% in certain DESs [133]. Although the activity and stability of some laccases were decreased in some DESs [134], their biocompatibility with DESs could be improved by protein engineering strategies, such as direct evolution. It is a rational speculation given that the laccase activity in ionic liquids can be improved by direct evolution [135] and that DESs resemble ionic liquids. Further protein engineering studies should be conducted to verify this speculation. Laccase can also degrade lignin into aromatic chemicals in aqueous ionic liquids [136]. Thus, it is reasonable to speculate that laccase can oxidatively degrade lignin into value-added compounds when using DESs as the reaction medium. Additionally, considering that DESs show excellent capacity for lignin fractionation from biomass, a consolidated process can be envisioned by conducting lignin fractionation and further conversion in the same DES by laccase oxidation. Similarly, the biocompatibility of other lignin-degrading enzymes (such as laccase, MnP, VP, and LigP) with DESs should be investigated to explore more biological routes for lignin depolymerization. In the future, substantial efforts should be made to promote the biological conversion of lignin as being time-saving and efficient.

6.2.2. Thermochemical route

Because of the aromatic structure of lignin, it is a potential precursor for the preparation of high value-added materials or chemicals via thermochemical conversion. Based on whether the catalyst is used or not, the thermochemical conversion of lignin can be classified as a non-catalytic or catalytic thermochemical transformation [100].

Non-catalytic thermochemical transformations. Pyrolysis and gasification are two main thermochemical transformation processes for lignin valorization. The pyrolysis process consists of heating lignin under an inert atmosphere at elevated temperatures (ranging from 150 to 700 °C). Lignin is later condensed into bio-oil and biochar as the major products [137]. The bio-oil is mainly composed of phenols, ketones, aldehydes, ethers, alcohols, and carboxylic acids, and can be further upgraded into other value-added chemicals. The biochar has many practical applications and is considered as a solid fuel source [138].

DES lignin has also shown impressive performance in lignin pyrolysis. For example, the DES lignin extracted by acidified ChCl:EG produced considerable amounts of phenolic monomers after pyrolysis [25,27]. Based on the pyrolysis performance, suitable downstream applications were proposed for the varying DES lignin obtained by pyrolysis using different DESs. Lignin fractionated by ChCl:FA was recommended for use as a carbon source (e.g., biochar) due to the high percentage of pyrolysis char residue, while lignin extracted by ChCl:LA was suitable for smaller-scale chemical production because its main pyrolysis products were volatile aromatic chemicals [86]. Compared with pyrolysis, gasification is normally conducted at even higher temperatures (>800 °C) under a reactive atmosphere (air, oxygen, or steam) to yield synthetic gas (such as CO, H_2, and CH_4) as the major product. These synthetic gases can further be converted into fuels or chemical products [137]. The TG-FTIR analysis showed that DES lignin (especially for ChCl:LA-fractionated) released high concentrations of gaseous products. This finding indicates its potential to serve as feedstock in gasification to produce syngas [86].

Catalytic thermochemical transformations. Catalytic thermochemical transformations mainly involve three pathways: (1) catalytic cracking/hydrolysis (into bio-oil or biochar), (2) catalytic reduction (into phenol or benzene, toluene and xylene, or BTX) via hydrogenolysis, hydrogenation, or hydrodeoxygenation, and (3) catalytic oxidation (into vanillin, acetovanillone, or acids) [137].

Lignin bio-oil is not suitable for direct use due to its high instability, corrosive nature, low heating value, and low incompatibility with petroleum fuels. Therefore, the stabilization and upgrading of lignin bio-oil are necessary to optimize its energy density and molecular weight, as well as to reduce the oxygen content [45]. The removal of oxygen atoms in lignin bio-oils (by removal of the methoxy and hydroxyl group, for example) can be triggered through direct C–O hydrogenolysis, C═C hydrogenation, and hydrodeoxygenation [139]. A possible valorization pathway of DES lignin toward value-added low-molecular weight phenolic compounds has been demonstrated [93]. More recently, a one-pot procedure consisting of extraction and depolymerization of lignin by DES has been developed [140]. This strategy offers a step-efficient protocol for producing a single and valuable catechol compound from lignocellulosic biomass.

The catalytic oxidation of lignin mainly produces more complex and valuable aromatic compounds with active functional groups. These functional groups include but are not limited to alcohol groups, aldehyde groups or carboxylic acid moieties. They offer many functionalization possibilities [141]. The resulted products normally are methoxy-phenolic aldehydes (vanillin, syringaldehyde, and *p*-hydroxy-benzaldehyde), carboxylic acids (vanillic acid and syringic acid), and ketones (acetovanillone and acetosyringone). A recent work reported an effective strategy for the catalytic conversion of lignin into versatile aromatic compounds. The H units of lignin were selectively catalyzed into methyl *p*-hydroxycinnamate as the sole product, with high yield and selectivity [142]. Two recent studies showed that lignin depolymerization could be accomplished through the solvolytic action of DESs (ChCl:LA or ChCl:OA) under relatively mild temperatures (80–140 °C). This depolymerization directly led to depolymerized lignin oil rich in value-added phenolic diketones [30,98]. Although no exogenous catalyst was added, the acidic environment of a DES might trigger the acid-catalyzed depolymerization of lignin. Significantly, depolymerized lignin products in DES systems contain unique diketone compounds that have not been observed after other types of acid pretreatment until now.

6.2.3. Electrochemical route

Lignin can be selectively electrochemically depolymerized into versatile aromatic compounds, such as vanillin [44] and acetovanillone [11]. Electrochemical depolymerization uses renewable electricity sources and is free from toxic and/or expensive oxidizers. This depolymerization route is green, inexpensive, and highly sustainable.

Currently, a feasible electrochemical strategy for lignin depolymerization includes reductive pathways and oxidative pathways [45]. In the reductive pathway, H_2 is normally used as the reductive agent for selective hydrogenolysis of lignin into phenols, while in the oxidative pathway, H_2O_2, O_2 and their resulting reactive oxygen species ($HO^{\bullet}$, $O_2^{-\bullet}$, or $HOO^{\bullet}$) play pivotal roles in the electrochemical depolymerization of lignin.

Generally, three operation modes of an electrode can be considered in most electrochemical depolymerization procedures for lignin (illustrated in Fig. 8) [143]. A crucial point is a working electrolyte/reaction medium that sufficiently dissolves lignin while ensuring satisfactory conductivity for electrochemical reaction. DESs are suitable candidates due to their satisfactory lignin solubility and electric conductivity. Di Marino et al. carried out pioneering work demonstrating that ChCl:EG dissolved lignin and served as an excellent medium for further electrochemical oxidative depolymerization of lignin into low-molecular-weight aromatic compounds [44,144]. The authors achieved efficient depolymerization of lignin through simultaneous fractionation of degraded products by *in situ* extraction, which prevented their over-oxidation [144]. Recently, a novel catalytic oxidation system was established that consisted of a copper catalyst in oxygenated ChCl:methanol that converted lignin into acetovanillone and acetic acid with high yield and selectivity [11].

Given the ability of DESs to efficiently fractionate lignin from biomass, it is highly conceivable to develop a consolidated strategy conducting lignin fractionation and further conversion in the same DESs by electrochemical oxidation. This consolidated process will eliminate steps in lignin recovery and reduce waste generation, which thus results in greater economic feasibility for lignin valorization [145]. Despite of all these advantages, some challenges still exist in such a strategy at present: (1) the lack of understanding regarding the structure-processing relationship for DESs-based electrochemical depolymerization; (2) the underdevelopment of catalysts/electrodes that are compatible and active in DES systems; and (3) the unoptimized reaction conditions (e.g., oxidation potential) which result in products with low yield and selectivity. More efforts are required to advance the electrochemical depolymerization of lignin in DES systems.

7. Future perspectives

DESs are an emerging class of green solvents that hold great promise for lignin fractionation and valorization. Although significant progress has been made in the past few years, several issues with respect to the fundamental mechanisms and technical challenges require rigorous investigations for future works.

Fundamental knowledge on how DES interacts with lignin during fractionation, dissolution, and valorization should be established. An in-depth investigation is needed on the structure-function relationship of DESs during fractionation to increase efficiency is needed. The underlying mechanisms of lignin dissolution and fractionation should be elucidated at a molecular level. More comprehensive research is required to understand the intrinsic mechanism and structural transformation changes of lignin during valorization. It is well-known that the addition of the correct amount of water will reduce the viscosity of DESs and contribute to applications. However, how much water can be added to DES without disrupting the solvent's intermolecular bonding structure is non-conclusive. Data collection on how water addition affects the DES molecular framework and its interaction with lignin would provide insightful information to facilitate the application of aqueous DESs. These insights will lay a foundation for the design, selection, and application of DESs.

In view of the tremendous amount of DES varieties, time- and resource-efficient screening approaches are required. At present, a vast majority of the studies involve the use of acidic DES. Other type of DESs, such as type V DESs, also show great potential in lignin fractionation and valorization. It is recommended to diversify the types of DESs adopted in lignin fractionation and valorization to fully explore the potential of DESs. Despite the direct design of DESs according to their intended purpose, which is promising, this strategy should be based on sound knowledge regrading the structure-function relationships common to DESs. Along with constantly increasing studies on DESs and applications, the available data on DESs will exponentially grow. Computational approaches such as machine learning, in which massive data sets are manipulated, are potent strategies to guide the selection of DESs.

More efforts are required to develop effective DESs recycling technologies. Changes in DES constituents, DES degradation, unexpected interactions with biomass-derived impurities that influence DES properties, and slightly unstable lignin structures and properties are found during the recyclable usage of DESs. The chemical structures and fundamental properties of DESs, the elimination of unexpected interactions, as well as the exploration of innovative technologies to remove impurities should be further investigated to increase the overall economic efficiency of the DESs-based biorefinery.

Detailed studies on the structures and properties of DESs-extracted lignin and comparison with other technical lignin are necessary. The possibility of applying DES lignin to existing lignin upgrading schemes needs to be evaluated, and the possible applications that have yet to be accomplished with the existing technical lignin should be explored. Additionally, DES lignin with tailored structures and properties can be achieved by adjusting DESs and process conditions. Currently, multiple types of upgraded lignin products such as aromatic compounds, bio-oil, biochar, nanoparticles, and smart materials have been successfully produced from the DESs-fractionated lignin in several works. However, the selectivity and yield of these upgrading routes greatly varied. Strategies to increase the usage of intended products will also be a great research direction in future works. Besides, it will also be worthwhile to explore more varieties of higher value lignin-derived commodity chemicals or materials to be converted from DESs-fractionated lignin to increase the process revenue.

A consolidated strategy can be envisioned by integrating lignin fractionation and further upgrading by the use of renewable DESs into a one-pot process. This strategy maximizes the utilization of biomass and minimizes the waste and use of chemicals and catalysts, and also creates opportunities toward a sustainable and circular bio-economy. Nevertheless, some technical challenges should be overcome. Many factors must be taken into consideration, including the type of biomass and its properties, reactor types, batch or continuous operations, recycling strategies, recovery and purification of product, and elimination of side-effects, as well as the development of DES-compatible catalysts. Presently, catalysts, enzymes, or electrodes that are compatible and active in DES environments remain under-exploited. In the future work, the development of DES-compatible catalysts to fractionate and valorize lignin with high selectivity and yield should be carried out. Exploration of new enzymes or modification of existing enzymes that are compatible with DESs through protein engineering is also highly encouraged. Furthermore, extensive life-cycle assessment and techno-economic analysis of consolidated strategies are also pressing research topics to further evaluate the feasibilities of lignin fractionation and valorization.

8. Conclusions

This review summarized the recent advances in lignin fractionation and valorization using DESs in view of newly emerging type V DESs, the underlying mechanisms to intensify strategies with DES treatment, the structure and properties of DES lignin, the mechanism of lignin dissolution in DESs and influencing factors, and the assistance of computational approaches in understanding the interactions between DESs and lignin. DESs have manifested promising performances in lignin fractionation and valorization. The employment of DESs will be a significant approach to reduce the dependency on non-renewable resources and move towards sustainable biorefineries. Despite significant progress that has been made, additional theoretical and engineering efforts are

required to understand the interactions between DESs and lignin at a molecular level, establish effective DESs recycling technologies, and developing more economically feasible lignin biorefinery strategies. With further research efforts, tunable DES systems for tailor-made lignin properties for cost-effective valorization will be expected in the near future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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